Conditional Latent-Prior Generative Compressed Sensing for Petrophysical Target-Log Reconstruction

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Abstract

Dense well logs provide continuous indirect observations of subsurface properties, whereas target-property curves may be available only at selected depths. Direct supervised models can combine dense logs with sparse target observations, but they typically use the sparse entries as input features and do not require the reconstruction to honor them as measurements. We introduce Conditional Latent-Prior Compressed Sensing with Generative Models (CLP-CSGM). A decoder trained on windows of the target curve supplies the windows the reconstruction may use. The dense logs set the starting point of the search for that window. At inference, the decoded window is adjusted so that it matches the sparse measurements. A conditional local-recovery bound separates decoder error, prior mismatch, measurement noise, and local separability. It does not prove that the measurement operator satisfies S-REC on the trained decoder. We evaluate CLP-CSGM on public Dutch North Sea well-log data reorganized into a held-out-well clay-volume benchmark and a single-well porosity stress test with weak dense

context. The F3 evaluation uses four wells, with F02-1, F03-2, and F06-1 for model development and F03-4 held out; the results are not presented as operational multi-well transfer. The clay-volume target is obtained from gamma ray by the Larionov relation, the single-well porosity target is processed and density-derived, and the sparse observations are coordinate subsamples of these same processed curves. The benchmarks therefore test reconstruction of processed target logs rather than recovery against independently acquired core or laboratory measurements. A limited-sample permeability case provides supplementary evidence. With a linear ridge map as the reference for that starting point, the comparison is a conditional advantage, not overall superiority. CLP-CSGM has lower mean RMSE than the direct autoencoder baseline in seven of nine cross-well cells, with the gains concentrated where training support is smallest (steps 16 and 32). The direct autoencoder has lower mean RMSE in the two remaining cells, both at step 8 (the largest training pool) for $\rho = 0.05$ and $\rho = 0.10$. The unweighted mean of the nine cell-level RMSE values decreases from 0.0350 to 0.0286, an overall reduction of about 18%. In the GR-only porosity benchmark, CLP-CSGM has the lowest mean RMSE at the two sparsest ratios, including 0.016 versus 0.018 at $\rho = 0.20$, whereas the direct autoencoder has lower mean RMSE at $\rho = 0.40$ and $\rho = 0.60$. That GR-only comparison is a controlled sparse-assimilation test under intentionally incomplete context: gamma ray is the only dense input, and the depth profile follows the broad trend of the processed porosity curve. It is not evidence of general porosity reconstruction performance. Dropping either the dense-log starting point or the match to the sparse measurements raises the error in most cells. Which of the two matters more changes with the number of training windows and with how many depths are measured. The map from dense logs to the starting point is also dataset-dependent: a multilayer perceptron gives lower mean RMSE on the cross-well grid, whereas ridge gives lower mean RMSE when gamma ray is the only dense log. On these processed-log tests, CLP-CSGM complements direct regression when training windows are few, the test well differs from the training wells, or the dense logs are incomplete.

Keywords: compressed sensing; generative model; inverse problem; well-log reconstruction; petrophysics; sparse assimilation

Computer Code Availability

CLP-CSGM (Conditional Latent-Prior Compressed Sensing with Generative Models) was developed by Romulo Brito da Silva, LENEP, North Fluminense State University, Rodovia Amaral Peixoto, km 163, Imboassica, Macae, RJ 27925-310, Brazil (contact: `romulobrito@lenep.uenf.br` ; telephone: +55 21 99101-3308). First publicly released in 2026, it is implemented in Python 3.10 using NumPy 1.26, pandas 2.3, Matplotlib 3.7, scikit-learn 1.6, PyTorch 2.6, and PyWavelets 1.4. The source repository occupies approximately 10 MB, excluding datasets and experiment outputs, and runs on a multi-core CPU with at least 8 GB of RAM; CUDA is optional.

Source code, installation instructions, dependencies, benchmark scripts, and documentation are available at `https://github.com/romulobrito/CLP--CSGM` under the MIT License. The repository covers the cross-well clay-volume and F03-4 porosity benchmarks, structural ablations, latent-prior sensitivity, and supplementary diagnostics. Processed well-log tables are not redistributed; the README specifies the required inputs and reproduction steps.

Data availability. The F3-derived benchmarks follow Zhang et al. [2025]; users must obtain the tabulated inputs from the cited releases or portals under the applicable licenses. The Lapa dataset follows Alvarez et al. [2025], whose underlying field data cannot be shared publicly. Supplementary analyses therefore require locally authorized inputs.

1 Introduction

Petrophysical target-log reconstruction builds a continuous property profile in a target well. Dense wireline logs run along the well, but they measure the target property only indirectly [Ellis and Singer, 2007, Asquith and Krygowski, 2004, Wang et al., 2024, Mondal and Singh, 2022, Sun et al., 2026]. A few depths already carry a value of that property. The task is to reconstruct the remaining depths while keeping those known values.

Supervised regressors learn a nonlinear map from logs to the target property. When the sparse observations are available, they are commonly joined to the dense logs and used as extra inputs [Bishop, 2006, Ahmadi and Chen, 2019, Wood, 2020, Pan et al., 2023, Mondal and Singh, 2022, Koray et al., 2024, Matinkia et al., 2023, Abdel Azim and Aljehani, 2022, Shi et al., 2024, He et al.,

2025, Markus et al., 2026]. After training, that map is not required to reproduce the observations it was given. In this work those observations stay measurements: the reconstructed window has to match them.

Recent studies complete a missing log interval, translate a geomodel property map, or regress porosity and shale volume from the log suite [Koeshidayatullah et al., 2026, Al-Fakih et al., 2026, 2025, 2023]. Each of those predictors returns the missing property from a trained map. Section 2 compares those trained maps with the measurement constraint used here.

Enforcing that match still requires a representation of the target window. A fixed sparsity basis is a weak description of this window when the logs change from one well to another [Donoho, 2006, Candes et al., 2006a,b, Bora et al., 2017]. Conditional Latent-Prior CSGM (CLP-CSGM) learns which target windows the decoder can produce and lets the dense logs set the starting point of the search. At inference the search stays near that point, and the decoded window must agree with the sparse measurements. Section 3.3 states this step. Ridge is the primary reference for the map from dense logs to that starting point, because it is a linear low-capacity map, not because it is best in every regime. An MLP is tested with the same pipeline.

The contributions are threefold. First, the dense logs set the starting point and the sparse values stay measurements that the reconstructed window must match. Second, the objective has a MAP/Tikhonov reading and a local error bound that separates decoder error, a poor starting point, noise, and whether nearby windows can be told apart from the measurements. Third, the tests use three development wells and hold out F03-4. They compare direct baselines, versions that drop the starting point or the measurement match, and a ridge map against a multilayer perceptron for the starting point. That local bound separates those error sources under stated assumptions. It is not a verified S-REC guarantee for the trained decoder.

2 Related work

2.1 Petrophysical log prediction

Well logs provide dense but indirect measurements of rock and fluid properties, and data-driven methods are widely used to predict petrophysical targets [Ellis and Singer, 2007, Asquith and Krygowski, 2004, Ahmadi and Chen, 2019, Wood, 2020]. Recent studies address three neighboring

tasks.

Log completion reconstructs missing intervals of a wireline curve from that curve. Sequence-based GANs generate synthetic logs and fill gaps in gamma ray, sonic, neutron porosity, deep resistivity, and bulk density [Al-Fakih et al., 2025]. A fine-tuned time-series foundation model forecasts the next interval of the same suite and flags anomalies, including a zero-shot check in a second basin [Koeshidayatullah et al., 2026]. The unknown is a missing piece of the log, and the trained predictor fills it in one pass.

Geomodel translation reconstructs a reservoir-property image from another image. A Pix2Pix conditional GAN maps two-dimensional slices of facies, porosity, permeability, and water saturation, taking either a masked image of the same property or a different property as generator input [Al-Fakih et al., 2026]. Agreement with the supplied image is learned in training.

Direct property regression maps wireline logs to a petrophysical target. ANN and random-forest models predict total porosity, effective porosity, and shale volume from logs that include neutron porosity [Al-Fakih et al., 2023]. Missing log values are treated in preprocessing. At inference the model returns the property from the logs, and target values act as training labels.

CLP-CSGM recovers a window of a target property when dense logs are available along the window and only sparse observations of that property are in hand. The dense logs set the starting point $\boldsymbol{z}_0 = h(\boldsymbol{u})$ of the search. The sparse values are measurements $\boldsymbol{b}$. After training, the code is adjusted so that the decoded window matches $\boldsymbol{b}$ at the observed depths. Log completion and geomodel translation fill a curve or a map from context. Direct regression predicts the property from the logs. CLP-CSGM keeps an explicit match to those measurements at inference. The difference is which quantity is unknown. In log completion, geomodel translation, and direct regression, the missing property values are what the trained model must produce. In CLP-CSGM those sparse values are already known measurements, and the unknown is the remainder of the target window, which the inference step is required to keep consistent with them.

The experiments use the processed F3 block data of Zhang et al. [2025], who combine supervised prediction with Gardner and Larionov relations and SHAP-based interpretation. We use these logs to study reconstruction from dense context while enforcing explicit consistency with sparse target-well observations.

2.2 Classical and generative compressed sensing

Classical compressed sensing combines an explicit measurement operator with a sparsity or compressibility prior [Donoho, 2006, Candes et al., 2006a,b]. Here, M_Ω is a coordinate restriction selecting observed target depths. A fixed transform-domain prior may be restrictive for cross-well petrophysical windows, whose representation can vary across wells. Restriction operators are also common in seismic, ultrasonic, and sparse-sensor reconstruction [Herrmann and Hennenfent, 2008, Hennenfent and Herrmann, 2008, de Assis et al., 2022, Muir and Zhan, 2021, da Silva et al., 2025].

CSGM replaces fixed-basis sparsity with the range $\mathrm{Im}(G)$ of a learned generator or decoder [Bora et al., 2017]. Reconstruction searches for a latent code $\boldsymbol{z}$ whose decoded signal matches the measurements. Recovery properties and limitations of this learned-range prior have been studied under set-restricted conditions and related generative formulations [Bora et al., 2017, Kamath et al., 2020, Asim et al., 2020]. CLP-CSGM retains this decoder-range view and adds dense-log conditioning of the latent search.

2.3 Conditional generative inverse problems and positioning

Conditional approaches differ in where side information and measurements enter. Direct conditional predictors map side information, alone or with observations, to an output or latent code [Mirza and Osindero, 2014, Sohn et al., 2015], while measurement-conditional CSGM conditions the generator directly on compressed measurements and their sampling pattern [Kim et al., 2021]. Other formulations retain an explicit inverse problem: side information may regularize a measurement-consistent reconstruction [Ongie et al., 2020, Lucas et al., 2018], approximate or sample inverse posteriors [Adler and Öktem, 2018, Altekrüger et al., 2024, Bendel et al., 2024], or use score/diffusion models for inverse reconstruction [Dasgupta et al., 2024, Jacobsen et al., 2025].

CLP-CSGM belongs to this explicit conditional inverse-reconstruction setting. Dense logs set the starting point $\boldsymbol{z}_0 = h(\boldsymbol{u})$ before the search, and the sparse target values enter only through the difference between the decoded window and those measurements. That starting point stays a penalty during the search. The experiments therefore compare a direct map from $[\boldsymbol{u}, \boldsymbol{b}]$ to the window, a reconstruction that uses only the starting point, and a reconstruction that uses only the measurements. Measurement-conditional generators and posterior-sampling architectures remain

outside the empirical scope.

3 Method

CLP-CSGM uses three pieces. A decoder, trained on windows of the target curve, supplies the windows the reconstruction may use. A map from the dense logs returns the starting code of the search. At inference the search stays near that code, and the decoded window must match the sparse measurements. This section defines the measurement model, the objective, the comparisons that drop one piece, and the error bound.

3.1 Problem setup

Let $\boldsymbol{y} \in \mathbb{R}^N$ denote a target-property window of N depth samples and let $\boldsymbol{u} \in \mathbb{R}^p$ denote the vectorized dense-log channels over the same interval. Target values are observed at the ordered coordinate set $\Omega = \{\omega_1, \ldots, \omega_m\} \subset \{1, \ldots, N\}$, where $1 \leq m < N$ and the measurement ratio is $\rho = m/N$.

The coordinate-restriction operator $M_\Omega : \mathbb{R}^N \to \mathbb{R}^m$ is represented by

$$M_\Omega = \begin{bmatrix} \boldsymbol{e}_{\omega_1}^\top \\ \vdots \\ \boldsymbol{e}_{\omega_m}^\top \end{bmatrix} \in \mathbb{R}^{m \times N}, \qquad M_\Omega \boldsymbol{y} = \begin{bmatrix} y_{\omega_1} \\ \vdots \\ y_{\omega_m} \end{bmatrix}, \tag{1}$$

where $\boldsymbol{e}_i$ is the i-th canonical basis vector of $\mathbb{R}^N$. Coordinate restriction is widely used to represent incomplete seismic, ultrasonic, and sparse-sensor observations [Herrmann and Hennenfent, 2008, de Assis et al., 2022, da Silva et al., 2025].

The sparse target measurements satisfy

$$\boldsymbol{b} = M_\Omega \boldsymbol{y} + \boldsymbol{\eta}, \qquad \boldsymbol{b}, \boldsymbol{\eta} \in \mathbb{R}^m, \tag{2}$$

where $\boldsymbol{\eta}$ is additive noise. Under the Gaussian model, $\boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{0}, \sigma_\eta^2 I_m)$; the noiseless case has $\boldsymbol{\eta} = \boldsymbol{0}$.

The dense logs and the sparse measurements do different jobs. The dense-log vector $\boldsymbol{u}$ sets the starting code $\boldsymbol{z}_0 = h(\boldsymbol{u})$. The measurements $\boldsymbol{b}$ enter only as the mismatch $\|M_\Omega G(\boldsymbol{z}) - \boldsymbol{b}\|_2^2$. The

reconstruction therefore starts from the dense logs and is then required to agree with the sparse target values.

Standardization and modeling assumptions. All preprocessing statistics are computed from training windows only. Let $\boldsymbol{\mu}_y, \boldsymbol{s}_y$ and $\boldsymbol{\mu}_u, \boldsymbol{s}_u$ be the componentwise training means and standard deviations, with $D_y = \mathrm{diag}(\boldsymbol{s}_y)$ and $D_u = \mathrm{diag}(\boldsymbol{s}_u)$. We use $\widetilde{\boldsymbol{y}} = D_y^{-1}(\boldsymbol{y} - \boldsymbol{\mu}_y)$ and $\widetilde{\boldsymbol{u}} = D_u^{-1}(\boldsymbol{u} - \boldsymbol{\mu}_u)$. Sparse measurements are standardized with $D_\Omega = M_\Omega D_y M_\Omega^\top$ and $\widetilde{\boldsymbol{b}} = D_\Omega^{-1}(\boldsymbol{b} - M_\Omega \boldsymbol{\mu}_y)$; near-zero standard deviations use the implementation safeguard. Predictions return to the original scale via

$$\widehat{\boldsymbol{y}} = \boldsymbol{\mu}_y + D_y \widehat{\widetilde{\boldsymbol{y}}}. \tag{3}$$

Subsequent equations omit tildes. We assume that target windows are adequately approximated by $\mathcal{M} = \mathrm{Im}(G) \subset \mathbb{R}^N$, with $G : \mathbb{R}^k \to \mathbb{R}^N$ and $k < N$, and that $h : \mathbb{R}^p \to \mathbb{R}^k$ provides a useful anchor $\boldsymbol{z}_0 = h(\boldsymbol{u})$. These assumptions are assessed empirically through reconstruction, ablations, and prior sensitivity.

3.2 Learned model components

Offline training fits an autoencoder on standardized target windows and a latent-prior predictor on paired dense-log windows and encoded targets. All learned components remain frozen during target-well reconstruction.

Generative decoder. Let $E_{\theta_E} : \mathbb{R}^N \to \mathbb{R}^k$ and $G_{\theta_G} : \mathbb{R}^k \to \mathbb{R}^N$, with $k < N$, denote the encoder and decoder. Given standardized training targets $\{\widetilde{\boldsymbol{y}}_i\}_{i=1}^{n_{\mathrm{tr}}}$, their parameters are obtained from

$$\begin{aligned}(\widehat{\theta}_E, \widehat{\theta}_G) \in \underset{\theta_E, \theta_G}{\arg\min} \frac{1}{n_{\mathrm{tr}}} \sum_{i=1}^{n_{\mathrm{tr}}} \left\| \widetilde{\boldsymbol{y}}_i - G_{\theta_G}\left(E_{\theta_E}\left(\widetilde{\boldsymbol{y}}_i\right)\right) \right\|_2^2 \\ + \gamma_{\mathrm{AE}} \left(\|\theta_E\|_2^2 + \|\theta_G\|_2^2 \right),\end{aligned} \tag{4}$$

where $\gamma_{\mathrm{AE}} \geq 0$ is the regularization weight. After training, we write $E = E_{\widehat{\theta}_E}$ and $G = G_{\widehat{\theta}_G}$. The frozen decoder defines the learned target-window class

$$\mathcal{M} = \mathrm{Im}(G) = \left\{ G(\boldsymbol{z}) : \boldsymbol{z} \in \mathbb{R}^k \right\} \subseteq \mathbb{R}^N. \tag{5}$$

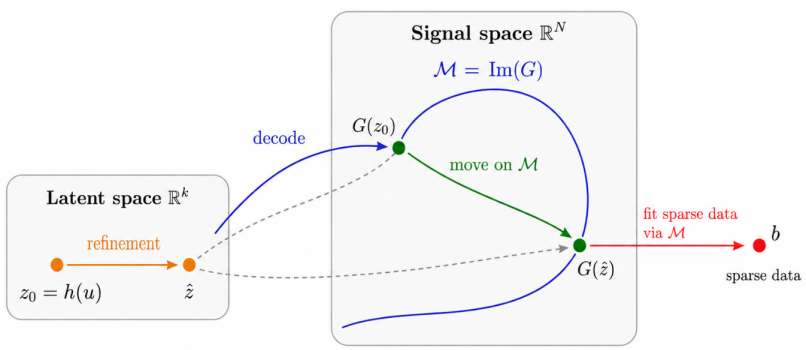


Figure 1: Geometry of the log-conditioned starting point. Dense logs determine $\boldsymbol{z}_0 = h(\boldsymbol{u})$, and test-time refinement moves the code to $\widehat{\boldsymbol{z}}$ so that $G(\widehat{\boldsymbol{z}}) \in \mathcal{M}$ better matches the sparse observations $\boldsymbol{b}$ through M_Ω.

Target-well reconstruction therefore optimizes only $\boldsymbol{z} \in \mathbb{R}^k$ and produces $\widehat{\boldsymbol{y}} = G(\widehat{\boldsymbol{z}}) \in \mathcal{M}$; it does not update the network parameters or search over arbitrary vectors in $\mathbb{R}^N$. The encoder is used to train the decoder and to build the codes on which the dense-log map is fit. Inference does not encode the unknown window. In the reported model, $k = 16$ is the dimension of that code. A reconstruction $G(E(\boldsymbol{y}))$ that is given the full target window checks whether the decoder can represent that window. It uses target values that inference does not have, so it is not one of the compared predictors. Appendix B.12 reports that check.

Log-conditioned latent prior. Each training target is encoded as $\boldsymbol{z}_i = E(\widetilde{\boldsymbol{y}}_i)$, and $h : \mathbb{R}^p \to \mathbb{R}^k$ is fitted from $\{(\widetilde{\boldsymbol{u}}_i, \boldsymbol{z}_i)\}_{i=1}^{n_{\mathrm{tr}}}$. For the Ridge variant, $h(\widetilde{\boldsymbol{u}}) = W\widetilde{\boldsymbol{u}} + \boldsymbol{c}$, with parameters

$$(\widehat{W}, \widehat{\boldsymbol{c}}) \in \arg\min_{W,\boldsymbol{c}} \left\{ \frac{1}{n_{\mathrm{tr}}} \sum_{i=1}^{n_{\mathrm{tr}}} \|\boldsymbol{z}_i - W\widetilde{\boldsymbol{u}}_i - \boldsymbol{c}\|_2^2 + \alpha \|W\|_F^2 \right\}, \tag{6}$$

where $\alpha \geq 0$ controls the Ridge penalty [Hoerl and Kennard, 1970]. After fitting, $h(\widetilde{\boldsymbol{u}}) = \widehat{W}\widetilde{\boldsymbol{u}} + \widehat{\boldsymbol{c}}$, and, suppressing tildes, the contextual anchor is $\boldsymbol{z}_0 = h(\boldsymbol{u}) \in \mathbb{R}^k$. The decoder G determines which target windows can be represented, and h uses dense-log context to center the latent search. CLP-CSGM is this scheme; the class of h is a configuration choice. Ridge is the primary reference because it is a linear map from the dense logs to the starting code, not because it is the best map in every test. An MLP variant of h is reported with the same pipeline in Section 6.4.

3.3 Test-time reconstruction

At inference time, G and h remain fixed, and only the latent code of the current target window is optimized. Given standardized dense logs $\widetilde{\boldsymbol{u}} \in \mathbb{R}^p$ and sparse target observations $\widetilde{\boldsymbol{b}} \in \mathbb{R}^m$, define $\boldsymbol{z}_0 = h(\widetilde{\boldsymbol{u}})$. CLP-CSGM computes

$$\begin{aligned} \widehat{\boldsymbol{z}}_{\text{num}} &\in \operatorname*{arg\,min}_{\boldsymbol{z}\in\mathbb{R}^k} \left\{ \left\| M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}} \right\|_2^2 + \lambda \left\| \boldsymbol{z} - \boldsymbol{z}_0 \right\|_2^2 \right\}, \\ \widehat{\boldsymbol{y}} &= \boldsymbol{\mu}_y + D_y G\left(\widehat{\boldsymbol{z}}_{\text{num}}\right). \end{aligned} \tag{7}$$

The first term enforces measurement consistency and the second regularizes latent search toward the anchor; $\lambda \geq 0$ is selected on validation windows for each measurement ratio and seed. Because G is nonlinear, the objective is generally nonconvex and is solved with Adam [Kingma and Ba, 2015]. The first run starts from $\boldsymbol{z}_0$; additional restarts use small Gaussian perturbations, and the lowest-objective candidate is retained. Thus $\widehat{\boldsymbol{z}}_{\text{num}}$ is a numerical solution, not a certified global minimizer.

Ablation variants. The *latent-prior-only* variant suppresses refinement and uses $\widehat{\boldsymbol{z}}_{\text{prior}} = \boldsymbol{z}_0$, followed by $\widehat{\boldsymbol{y}}_{\text{prior}} = \boldsymbol{\mu}_y + D_y G(\widehat{\boldsymbol{z}}_{\text{prior}})$. The *measurement-only CSGM* variant sets $\lambda = 0$ and computes

$$\widehat{\boldsymbol{z}}_{\text{meas}} \in \operatorname*{arg\,min}_{\boldsymbol{z}\in\mathbb{R}^k} \left\| M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}} \right\|_2^2. \tag{8}$$

The latent-prior-only variant uses only the dense-log starting point. The measurement-only variant uses only the sparse measurements and recovers measurement-only generative compressed sensing [Bora et al., 2017]. Comparison with the full objective in Eq. (7) shows what each piece contributes. Results are reported in Section 6.3.

3.4 Interpretation and theoretical context

The theoretical results required to interpret CLP-CSGM are stated below. Complete proofs appear in Appendix A.

Proposition 1 (MAP and Tikhonov interpretation)**.** *Suppressing tildes, assume* $p(\boldsymbol{b} \mid \boldsymbol{z}, \boldsymbol{u}) = p(\boldsymbol{b} \mid \boldsymbol{z})$ *and*

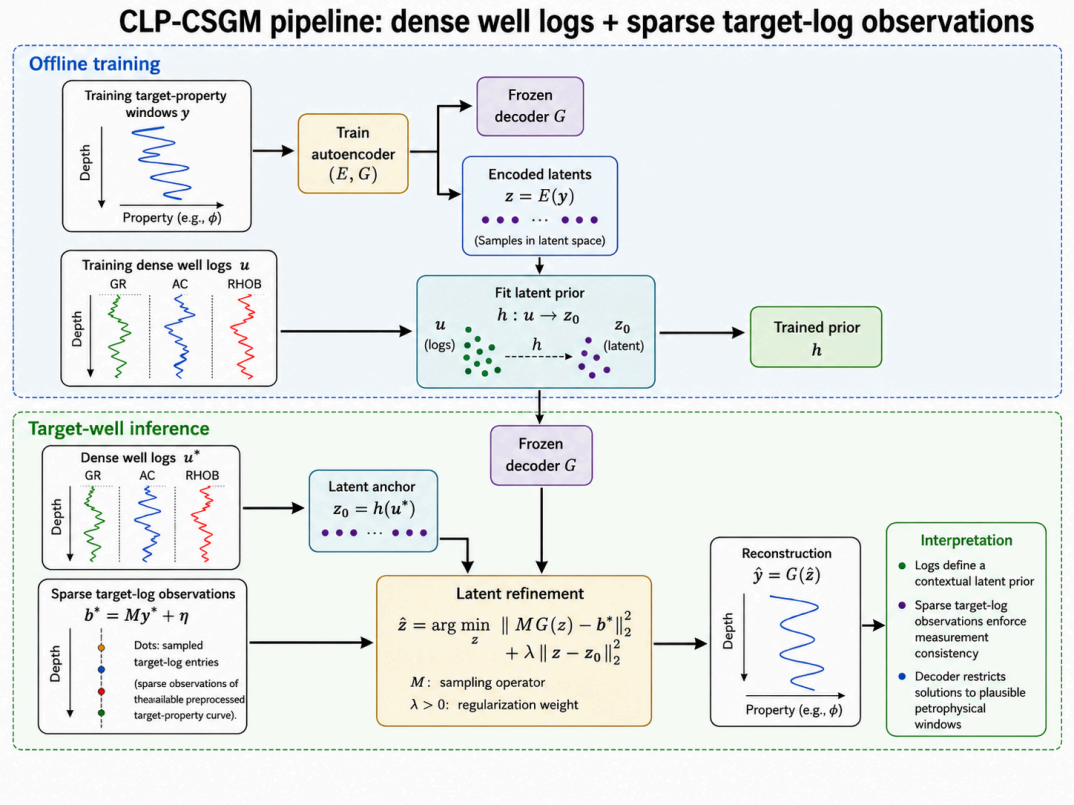


Figure 2: CLP-CSGM workflow. Offline, standardized target windows train the autoencoder (E, G), and paired dense-log windows fit the latent-prior predictor h. At inference, $\boldsymbol{u}_*$ determines $\boldsymbol{z}_0 = h(\boldsymbol{u}_*)$, sparse observations $\boldsymbol{b}_*$ constrain latent refinement through M_Ω, and the decoded result is returned to the original target scale. OpenAI's ChatGPT, using the GPT-5.5 model, was used to assist in generating the initial explanatory schematic and visual layout. The final figure was reviewed, edited, and validated by the authors.

$\boldsymbol{b} \mid \boldsymbol{z} \sim \mathcal{N}\left(M_\Omega G(\boldsymbol{z}), \sigma_\eta^2 I_m\right)$, *while* $\boldsymbol{z} \mid \boldsymbol{u} \sim \mathcal{N}\left(\boldsymbol{z}_0, \sigma_z^2 I_k\right)$, *where* $\boldsymbol{z}_0 = h(\boldsymbol{u})$, $\sigma_\eta > 0$, *and* $\sigma_z > 0$. *Then every MAP estimator satisfies*

$$\widehat{\boldsymbol{z}}_{\mathrm{MAP}} \in \underset{\boldsymbol{z}\in\mathbb{R}^k}{\arg\min} \left\{ \|M_\Omega G(\boldsymbol{z}) - \boldsymbol{b}\|_2^2 + \lambda \|\boldsymbol{z} - \boldsymbol{z}_0\|_2^2 \right\}, \qquad \lambda = \frac{\sigma_\eta^2}{\sigma_z^2}. \tag{9}$$

Thus, CLP-CSGM is a Tikhonov-regularized latent inverse problem. Its center is the log-conditioned anchor, not the origin [Tikhonov, 1963, Burger and Lucka, 2014]. The likelihood induces the data-fidelity term, while the conditional prior induces the latent penalty.

Remark 1 (Relation to classical CSGM). Setting $\lambda = 0$ recovers the measurement-only CSGM problem in Eq. (8) [Bora et al., 2017]. Increasing λ shifts the reconstruction toward the latent-prior estimate $G(\boldsymbol{z}_0)$ by penalizing departures from $\boldsymbol{z}_0 = h(\boldsymbol{u})$ more strongly.

Proposition 2 (Existence of a minimizer)**.** *Assume that $G : \mathbb{R}^k \to \mathbb{R}^N$ is continuous and $\lambda > 0$. For fixed M_Ω, $\widetilde{\boldsymbol{b}} \in \mathbb{R}^m$, and $\boldsymbol{z}_0 \in \mathbb{R}^k$, the objective $\mathcal{J}(\boldsymbol{z}) = \|M_\Omega G(\boldsymbol{z}) - \widetilde{\boldsymbol{b}}\|_2^2 + \lambda\|\boldsymbol{z} - \boldsymbol{z}_0\|_2^2$ is continuous and coercive. Consequently,*

$$\arg\min_{\boldsymbol{z}\in\mathbb{R}^k} \mathcal{J}(\boldsymbol{z}) \neq \varnothing. \tag{10}$$

For any global minimizer $\widehat{\boldsymbol{z}}$, $\mathcal{J}(\widehat{\boldsymbol{z}}) \leq \mathcal{J}(\boldsymbol{z}_0) = \|M_\Omega G(\boldsymbol{z}_0) - \widetilde{\boldsymbol{b}}\|_2^2$. Exact minimization therefore cannot worsen the optimized objective relative to the latent-prior-only candidate, although it does not guarantee lower error with respect to the unknown target. For $\widehat{\boldsymbol{z}}_{\text{num}}$, this comparison holds only when $\mathcal{J}(\widehat{\boldsymbol{z}}_{\text{num}}) \leq \mathcal{J}(\boldsymbol{z}_0)$.

Remark 2 (Representation error and decoder mismatch). Suppressing tildes when standardized coordinates are understood, every CLP-CSGM reconstruction satisfies $\widehat{\boldsymbol{y}} = G(\widehat{\boldsymbol{z}}) \in \mathcal{M}$, where $\mathcal{M} = \mathrm{Im}(G) \subseteq \mathbb{R}^N$. Consequently, for an unknown target window $\boldsymbol{y}^\star \in \mathbb{R}^N$, the error of any decoder-based reconstruction is bounded below by the oracle representation error

$$\mathrm{dist}\left(\boldsymbol{y}^\star, \mathcal{M}\right) := \inf_{\boldsymbol{y}'\in\mathcal{M}} \left\|\boldsymbol{y}^\star - \boldsymbol{y}'\right\|_2 = \inf_{\boldsymbol{z}\in\mathbb{R}^k} \left\|\boldsymbol{y}^\star - G(\boldsymbol{z})\right\|_2 . \tag{11}$$

Thus, even with noiseless observations and exact global optimization, latent refinement cannot remove mismatch between $\boldsymbol{y}^\star$ and the decoder range. This structural error is represented by $\boldsymbol{r}^\dagger$ in Proposition 3 and is distinct from the latent-prior mismatch $\|\boldsymbol{z}^\dagger - \boldsymbol{z}_0\|_2$.

An inaccurate anchor $\boldsymbol{z}_0 = h(\boldsymbol{u})$ may bias the search toward a suboptimal latent region. The data-fidelity term can partially correct this bias when the sparse observations are informative and λ does not dominate the objective. The decoder range determines what can be represented; the log-conditioned anchor determines where the latent search is centered.

Proposition 3 (Conditional local recovery interpretation)**.** *Suppressing tildes, let $\boldsymbol{y}^\star \in \mathbb{R}^N$ satisfy $\boldsymbol{b} = M_\Omega \boldsymbol{y}^\star + \boldsymbol{\eta}$, let $\boldsymbol{z}_0 = h(\boldsymbol{u})$, $\lambda > 0$, and let $\widehat{\boldsymbol{z}}$ be any global minimizer of Eq.* (7)*. For any reference code $\boldsymbol{z}^\dagger \in \mathbb{R}^k$, define*

$$\begin{aligned} \boldsymbol{r}^\dagger &= \boldsymbol{y}^\star - G(\boldsymbol{z}^\dagger), & \boldsymbol{e}^\dagger &= M_\Omega \boldsymbol{r}^\dagger + \boldsymbol{\eta}, \\ \Delta^\dagger &= \left(\|\boldsymbol{e}^\dagger\|_2^2 + \lambda\|\boldsymbol{z}^\dagger - \boldsymbol{z}_0\|_2^2\right)^{1/2}, & R^\dagger &= \frac{\Delta^\dagger}{\sqrt{\lambda}}. \end{aligned} \tag{12}$$

Let $\mathcal{M}_{R^\dagger} = G(B(\boldsymbol{z}_0, R^\dagger))$*, where* $B(\boldsymbol{z}_0, R^\dagger) = \left\{\boldsymbol{z} \in \mathbb{R}^k : \|\boldsymbol{z} - \boldsymbol{z}_0\|_2 \le R^\dagger\right\}$*. If, for some* $\gamma_R > 0$ *and* $\delta_R \ge 0$*,*

$$\|M_\Omega(\boldsymbol{y}_1 - \boldsymbol{y}_2)\|_2 \ge \gamma_R \|\boldsymbol{y}_1 - \boldsymbol{y}_2\|_2 - \delta_R, \qquad \forall\, \boldsymbol{y}_1, \boldsymbol{y}_2 \in \mathcal{M}_{R^\dagger}, \tag{13}$$

then

$$\|G(\widehat{\boldsymbol{z}}) - \boldsymbol{y}^\star\|_2 \le \|\boldsymbol{r}^\dagger\|_2 + \frac{\Delta^\dagger + \|\boldsymbol{e}^\dagger\|_2 + \delta_R}{\gamma_R}. \tag{14}$$

Proposition 3 is a sufficient conditional interpretation, not a verified guarantee for the trained decoder. The latent penalty localizes the S-REC condition around $\boldsymbol{z}_0$, while the bound separates representation error, measurement residual/noise, prior mismatch, and local separability, following the decoder-range view of generative compressed sensing [Candes et al., 2006b, Bora et al., 2017]. Coordinate restriction is not proved to satisfy S-REC on the trained range; this depends on local decoder secants. The proof, inexact-optimizer extension, and empirical separability diagnostics appear in Appendix A.3 and Appendix B.1.

4 Data and exploratory analysis

We evaluate two F3 settings. In the first, clay volume is reconstructed in a held-out well. In the second, porosity in one well is reconstructed from gamma ray alone.

4.1 Field settings and benchmark provenance

The experiments use the processed F3 well-log curves of Zhang et al. [2025]; no new field data were acquired. We use the FS2–FS6 intervals of F02-1, F03-2, F03-4, and F06-1. Sonic and gamma ray are available in all wells; density is measured in F02-1/F03-2 and predicted in F03-4/F06-1. Porosity is density-derived and V_c is Larionov-derived from gamma ray; further geological and processing details are given in the source study.

The cross-well benchmark reconstructs V_c from 30 dB noisy logs, using F02-1, F03-2, and F06-1 for development and F03-4 for testing. The single-well benchmark reconstructs F03-4 porosity from gamma ray alone. In both, $\boldsymbol{b}$ is obtained by coordinate subsampling of the processed target.

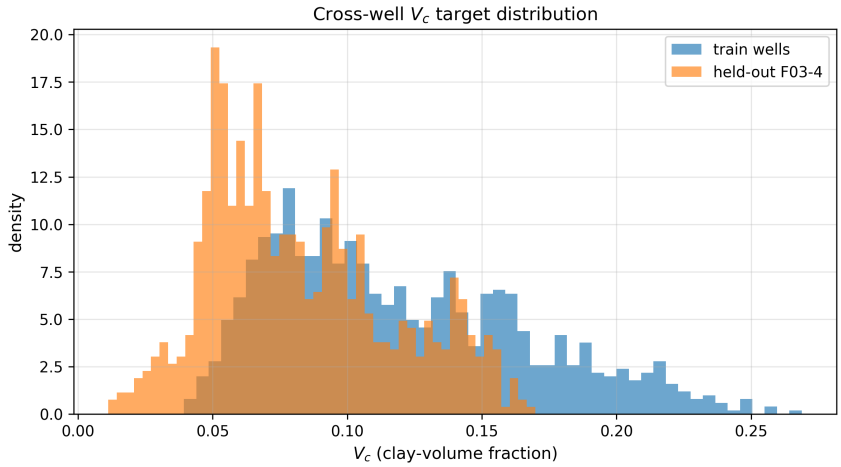


Figure 3: Empirical cross-well V_c distributions for the pooled training wells and the held-out F03-4 well. The held-out distribution is shifted toward lower clay-volume values while retaining partial overlap with the training distribution.

Appendix D provides a limited-sample Lapa permeability case based on Alvarez et al. [2025].

Table 1: Summary of the two F3 benchmarks. The cross-well target V_c is derived from gamma ray through the Larionov relation, whereas the single-well target is processed porosity.

dataset	well	target	n_rows	depth_min	depth_max	target_mean	target_std	target_min	target_max
cross_train	F02-1	V_c	165	1149.7056	1174.6992	0.1215	0.0407	0.0513	0.1934
cross_train	F03-2	V_c	811	951.9000	1073.4000	0.1184	0.0490	0.0393	0.2688
cross_train	F06-1	V_c	122	1133.1000	1151.2500	0.1223	0.0437	0.0427	0.2175
cross_test	F03-4	V_c	832	946.5000	1071.1500	0.0824	0.0341	0.0112	0.1699
real_f03_gr	F03-4	Porosity	1435	856.0500	1071.1500	0.2789	0.0142	0.2471	0.3264

Table 1 summarizes the two settings. Held-out F03-4 has mean $V_c = 0.0824$, versus about 0.12 in the training wells, and Fig. 3 shows a lower-valued but overlapping distribution. Larger sliding-window steps further reduce training support, whereas CLP-CSGM can still refine the held-out-well estimate using $\boldsymbol{b}$ at inference.

Channel-target correlations and input design. Table 2 reports descriptive Pearson correlations used only to guide input design. Clay volume in this benchmark is computed from gamma ray by the Larionov relation [Zhang et al., 2025]. The main cross-well input therefore leaves gamma ray out and uses sonic, density, acoustic impedance, and V_p. With gamma ray included, the model would read the log from which the target was computed. This is not the usual field interpretation, in which gamma ray is logged and clay volume is obtained from that relation directly. The gamma-ray index between the clean and shale endpoints is linear in gamma ray. The Larionov map from that index to clay volume is not a straight line. When gamma ray has been logged, that relation is the

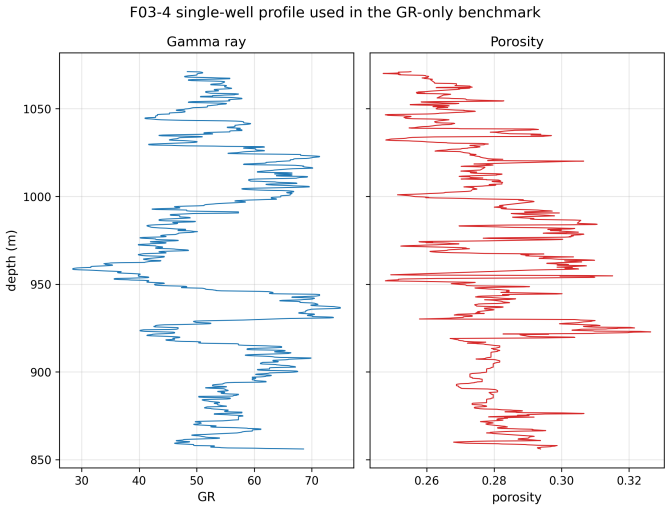


Figure 4: F03-4 depth profiles used in the GR-only porosity benchmark. Gamma ray is the only dense input, while porosity is reconstructed from the log-conditioned latent prior and sparse target observations.

calculation, and the model is not a substitute for it. Gamma ray is put back only in Appendix B.7, as a check on the effect of that channel.

Table 2: Pearson correlations between the available input channels and the target properties in the cross-well V_c and single-well F03-4 porosity benchmarks.

dataset	channel	target	pearson_r
cross_train	Sonic	V_c	-0.1642
cross_train	RHOB	V_c	0.0807
cross_train	GR	V_c	0.9733
cross_train	AI	V_c	0.1672
cross_train	V_p	V_c	0.1819
cross_test	Sonic	V_c	-0.1606
cross_test	RHOB	V_c	-0.0061
cross_test	GR	V_c	0.9772
cross_test	AI	V_c	0.1330
cross_test	V_p	V_c	0.1077
real_f03	AC	Porosity	0.9996
real_f03	GR	Porosity	-0.1759

In F03-4, sonic travel time follows the processed porosity almost one to one, whereas gamma ray is only weakly related to it. Sonic is left out for that reason. The density curve behind this porosity was itself predicted from sonic and gamma ray. The test that remains uses gamma ray as the only dense log and keeps a few values of the porosity curve as measurements. It is not a porosity interpretation from density, neutron, and sonic together (Fig. 4).

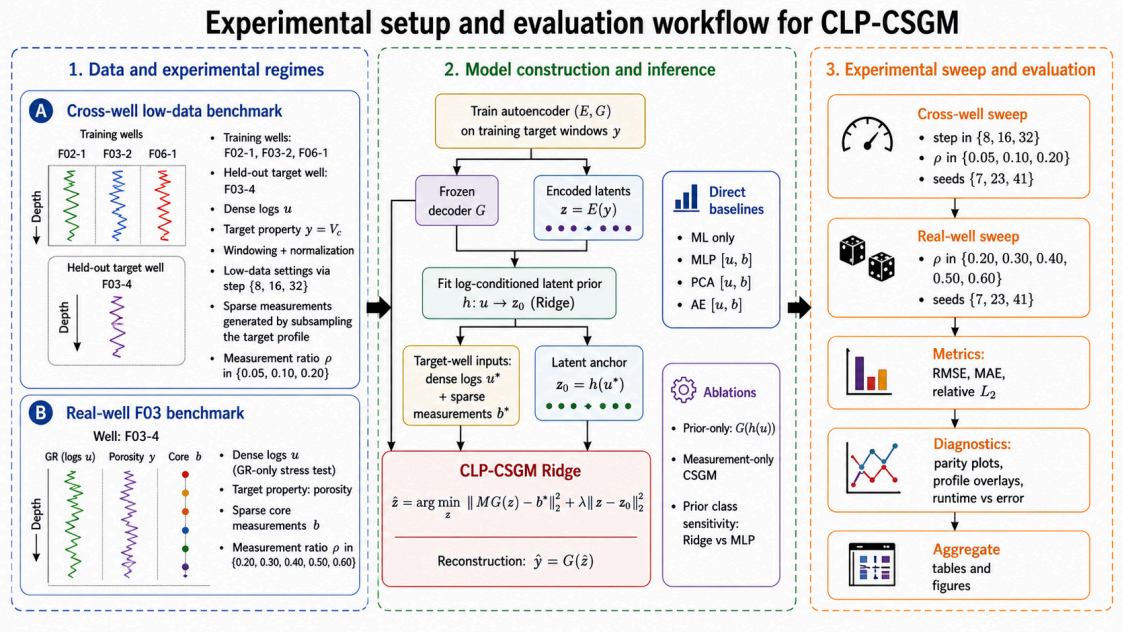


Figure 5: Experimental protocol. Offline training fits the decoder and the map from dense logs to the starting code. At test time that code is adjusted on the target well. Partitions, measurement ratios, seeds, and sparse-observation draws are shared across methods, which are evaluated using RMSE, MAE, and relative ℓ_2 error. OpenAI's ChatGPT, using the GPT-5.5 model, was used to assist in generating the initial explanatory schematic and visual layout. The final figure was reviewed, edited, and validated by the authors.

5 Experimental setup

The evaluation uses paired protocols for the cross-well low-data V_c benchmark and the single-well F03-4 GR-only porosity benchmark. Within each cell, all methods share the same partitions, measurement ratios, random seeds, sampled coordinates, and noise realizations.

5.1 Cross-well clay-volume (V_c) low-data benchmark

The cross-well benchmark reconstructs length-$N = 64$ processed V_c windows in the held-out F03-4 well. F02-1, F03-2, and F06-1 provide the model-development data, and the dense input $\boldsymbol{u}$ concatenates sonic, bulk density, acoustic impedance, and V_p. Reported cross-well performance refers to this single held-out well. As justified in Section 4, gamma ray is left out because clay volume was computed from that log [Zhang et al., 2025]. This is not the log suite for a well in which gamma ray is available and clay volume is obtained from it directly.

Sparse observations follow $\boldsymbol{b} = M_\Omega \boldsymbol{y} + \boldsymbol{\eta}$ at $\rho \in \{0.05, 0.10, 0.20\}$. Sliding-window steps $\{8, 16, 32\}$ control training support, and each step–ρ cell uses three paired seed realizations. This

is a controlled sparse-assimilation benchmark on a processed target curve, not recovery against an independently acquired measurement. Split and sampling details are given in Section 5.3; GR inclusion and the embargo-style boundary check are reported in Appendix B.7.

5.2 Single-well F03-4 GR-only porosity benchmark

The second benchmark reconstructs length-$N = 64$ windows of the processed density-derived F03-4 porosity curve using gamma ray as the only dense input. Acoustic travel time is excluded because it is nearly collinear with the target (Pearson $r \approx 0.9996$) and would provide a near-direct proxy. Density porosity is linear in bulk density, and a standard sonic porosity is linear in slowness. If those logs have been recorded, the processed porosity is already determined, and this test is not a substitute for those calculations. Gamma ray is then the only dense log, so the dense context is intentionally incomplete. The target follows the processing of Zhang et al. [2025], in which the density curve underlying F03-4 porosity is itself predicted from sonic and gamma ray.

Sparse observations are sampled at $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$. Depth is partitioned before window construction, preventing overlap among training, validation, and test windows. The benchmark is a controlled test with gamma ray as the only dense log. It is not a recommendation to interpret porosity from gamma ray alone, and its errors do not transfer to a suite that also includes density, neutron, and sonic.

The smaller Lapa pre-salt permeability case [Alvarez et al., 2025] is retained as supplementary evidence in Appendix D. Appendix E reports a separate single-well check on laboratory porosity. The direct AE has the lower mean held-out RMSE. Laboratory measurements correct the log-only anchor in every split, and that correction is enough where the test logs remain close to the anchor's training support. The check is not part of the main grid.

5.3 Training, validation, and test protocol

All preprocessing statistics and model choices are determined without test-target leakage. Training statistics are reused unchanged on validation and test windows. The training split fits the direct baselines, autoencoder (E, G), and latent-prior predictor h. Validation targets select λ and baseline-specific hyperparameters, and test targets are used only for final evaluation.

In the cross-well benchmark, F02-1, F03-2, and F06-1 form the development pool, F03-4 is

held out entirely for testing, and 10% of the development windows are reserved for validation. Steps $\{8, 16, 32\}$ produce approximately $\{104, 52, 27\}$ windows before this split. At step 32 the development pool is small; Appendix B.8 redraws that pool ten times with the measurement pattern held fixed. CLP-CSGM Ridge has lower RMSE than AE $[\boldsymbol{u}, \boldsymbol{b}]$ at every ρ and in all ten splits. Because the validation tail is selected after window construction, boundary windows may overlap when step $< N$; Appendix B.7 reports an embargo-style check that removes these overlap-prone windows.

For F03-4 porosity, depth samples are divided into contiguous training, validation, and test blocks using fractions 60%, 20%, and 20% before windows are constructed independently. This depth-first partition prevents overlap and preserves contiguous-depth generalization.

For each seed and measurement ratio ρ, the number of observed coordinates is

$$m = \max\{4, \mathrm{round}(\rho L)\}.$$

We use nearest-integer rounding, with half-integer ties resolved to the nearest even integer. One coordinate set $\Omega = \{\omega_1, \ldots, \omega_m\}$ is sampled for each (seed, ρ) cell and held fixed across all training, validation, and test windows in that cell. The set defines

$$\boldsymbol{b} = \begin{bmatrix} y_{\omega_1} & \cdots & y_{\omega_m} \end{bmatrix}^\top + \boldsymbol{\eta}. \tag{15}$$

Within each (seed, ρ) cell, all methods share the partitions, the fixed sampled coordinates, their ordering, and the noise realization. Direct baselines receive $[\boldsymbol{u}, \boldsymbol{b}] \in \mathbb{R}^{p+m}$, with only the measured target values in the ordering of M_Ω. Because Ω is fixed within the cell, the j-th entry of $\boldsymbol{b}$ consistently corresponds to coordinate ω_j across windows, making coordinate identity implicit in the stable input ordering. No zero-padded target, binary mask, or measurement matrix is added. CLP-CSGM uses M_Ω explicitly in $\|M_\Omega G(\boldsymbol{z}) - \boldsymbol{b}\|_2^2$. The protocol therefore compares direct fusion with operator-based assimilation under a fixed sampling pattern, giving the direct models a stable coordinate-to-input mapping within each cell. Appendix B.10 gives one direct AE the same measurements on the window coordinates plus a binary mask. That mask does not remove the CLP-CSGM advantage at steps 16 and 32; at step 8 the masked AE remains better at $\rho = 0.05$ and $\rho = 0.10$, and the mask

does not uniformly lower the direct AE error. This comparator is not part of the main grid.

Appendix B.11 redraws the coordinate set five times with the model seeds held fixed. Where training support is smallest (steps 16 and 32), CLP-CSGM Ridge keeps the lower RMSE in every draw. At step 8, where the development pool is largest, the ranking is less stable: AE $[\boldsymbol{u}, \boldsymbol{b}]$ stays ahead at $\rho = 0.05$ and on the majority of draws at $\rho = 0.10$, while CLP-CSGM stays ahead on the majority of draws at $\rho = 0.20$. The cell-level winner still matches the published ranking in all nine cells. That check is not part of the main grid.

Appendix B.12 adds direct linear and ridge maps from $[\boldsymbol{u}, \boldsymbol{b}]$ to the window, and depth interpolation of the sparse measurements, on this same cross-well grid. It also raises the measurement-noise standard deviation above the published value 0.01. Those comparators and that noise sweep stay outside the main grid.

Noise is applied after subsampling, with $\sigma_\eta = 0.01$ for cross-well V_c and $\sigma_\eta = 0.02$ for F03-4 porosity, independently across coordinates and windows.

5.4 Implementation details

Table 3 reports the architectures, optimization settings, and validation grids. The compact backbone uses latent dimension $k = 16$ and a decoder that remains frozen during target-well reconstruction.

For CLP-CSGM, the first latent-refinement run starts from $\boldsymbol{z}_0 = h(\boldsymbol{u})$, additional runs use small Gaussian perturbations around this anchor, and the candidate with the lowest objective is retained. The measurement-only ablation sets $\lambda = 0$ and starts from zero, whereas the prior-only variant returns $G(h(\boldsymbol{u}))$ without refinement. Predictions are returned to the original scale using Eq. (3).

5.5 Baselines and evaluation metrics

The direct baselines are ML only, which uses only $\boldsymbol{u}$; MLP $[\boldsymbol{u}, \boldsymbol{b}]$, which predicts the target directly; PCA $[\boldsymbol{u}, \boldsymbol{b}]$, which predicts target principal-component scores; and AE $[\boldsymbol{u}, \boldsymbol{b}]$, which predicts codes of a separately trained target autoencoder. CLP-CSGM Ridge is the proposed reference configuration. Prior-only Ridge and measurement-only CSGM are structural ablations, while the MLP latent prior is used only for sensitivity analysis.

Metrics are computed after inverse standardization on the original target scale. RMSE is the primary metric and selects λ on validation data; MAE and relative ℓ_2 error provide complementary

Table 3: Implementation settings for offline training, target-well latent refinement, and direct supervised baselines.

Phase	Module	Implementation
Offline training	Standardization	Training-set mean and standard deviation for u and y; reused unchanged on validation and test windows.
	Autoencoder (E, G)	MLP with one hidden layer (width 128) in encoder and decoder, latent dimension $k = 16$, ReLU activations, linear output; 200 epochs, Adam [Kingma and Ba, 2015] (10^{-3}), weight decay 10^{-5}, batch size 64.
	Latent prior $h(u)$	Ridge regressor with $\alpha = 1.0$ on standardized (u, z) pairs (main experiments). Sensitivity variant: MLP with two hidden layers (width 128), L_2 penalty 10^{-4}, learning rate 10^{-3}, max. 500 iterations, early stopping.
Target-well inference	Latent refinement	Adam [Kingma and Ba, 2015] for 400 iterations, learning rate 0.05, three restarts (anchor $z_0 = h(u)$ plus perturbed starts); λ selected on validation from $\{10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}, 3{\cdot}10^{-2}, 10^{-1}\}$.
	Reconstruction	$\hat{y} = G(\hat{z})$ after inverse standardization to the original target scale.
Direct baselines	$[u, b]$ regressors	ML only and MLP $[u, b]$: sklearn MLP with two hidden layers (width 128), max. 500 iterations, learning rate 10^{-3}, L_2 penalty 10^{-4}, early stopping. PCA $[u, b]$: PCA score grid $\{8, 16, 24, 32, 48, 64\}$ with the same MLP head. AE $[u, b]$: separate MLP autoencoder ($k = 32$, hidden width 128, 120 epochs) plus MLP from $[u, b]$ to latent codes.

summaries. Results are computed under the paired protocol for each seed and then summarized across seeds, with lower values indicating better reconstruction.

Uncertainty summaries. All seed-level intervals use the three fixed seeds $\{7, 23, 41\}$. Reported 95% half-widths are Student-t intervals with $\mathrm{df} = n_{\mathrm{seeds}} - 1 = 2$ (critical value $t_{2,0.975} \approx 4.303$), not normal $z = 1.96$ intervals. Where CLP-CSGM Ridge is compared with AE $[\boldsymbol{u}, \boldsymbol{b}]$, we also report the paired seed-wise difference

$$\Delta\mathrm{RMSE} = \mathrm{RMSE}_{\mathrm{AE}} - \mathrm{RMSE}_{\mathrm{CLP}},$$

together with its seed-level standard error and Student-t half-width. Positive ΔRMSE favors CLP-CSGM Ridge. With only three seeds, these intervals describe seed-level reproducibility within the paired protocol; their interpretation remains limited by the small seed panel. Two supplements widen that evidence and leave the three-seed tables as the main result. Appendix B.9 repeats the comparison on ten seeds. Each extra seed refits the models and draws a new coordinate set,

so training randomness and observation placement change together. None of those nine intervals include zero. The CLP-CSGM advantage remains where training support is smallest (steps 16 and 32); AE remains better at step 8 for $\rho = 0.05$ and $\rho = 0.10$. Appendix B.11 holds the same three model seeds and redraws only the coordinate set. The advantage at steps 16 and 32 holds in every draw, and the step-8 ranking is the less stable one. Overlapping windows inside a well are not counted as further independent samples.

6 Results

All comparisons follow the paired protocol of Section 5: within each (seed, ρ) cell, the methods share partitions, sampled coordinates, and noisy sparse observations. Method orderings in this section refer to the reported mean metrics across the paired seeds.

6.1 Cross-well low-data results

Table 4 compares CLP-CSGM Ridge with AE $[\boldsymbol{u}, \boldsymbol{b}]$, the direct baseline with the lowest unweighted mean RMSE across the grid. CLP-CSGM Ridge has lower mean RMSE in seven of the nine cells; the exceptions are step 8 with $\rho \in \{0.05, 0.10\}$. It has lower mean RMSE at every tested ratio for steps 16 and 32. Across the three tested step sizes, the descriptive RMSE gap shifts progressively in favor of CLP-CSGM as training support decreases. At step 32, the RMSE reductions relative to AE are approximately 32% at $\rho = 0.10$ and 57% at $\rho = 0.20$.

Across the complete grid, the unweighted mean of the nine cell-level mean RMSE values decreases from 0.03496 for AE $[\boldsymbol{u}, \boldsymbol{b}]$ to 0.02862 for CLP-CSGM Ridge, a reduction of approximately 18.1%. The ordering still depends on the amount of training data: the direct AE has lower mean RMSE in the two cells with the largest training pool, step 8 at $\rho = 0.05$ and $\rho = 0.10$.

Table 4: Cross-well V_c comparison between CLP-CSGM Ridge and AE $[\boldsymbol{u}, \boldsymbol{b}]$. Values are RMSE mean and standard deviation across seeds; a negative gap favors CLP-CSGM.

step	n_train_approx	rho	clp_csgm_rmse_mean	clp_csgm_rmse_std	ae_ub_rmse_mean	ae_ub_rmse_std	gap_vs_ae_pct
8	104	0.05000	0.04483	0.00524	0.03567	0.00100	25.69014
8	104	0.10000	0.03646	0.00194	0.03195	0.00208	14.10258
8	104	0.20000	0.02267	0.00381	0.02600	0.00164	-12.83374
16	52	0.05000	0.03344	0.00151	0.03712	0.00098	-9.91538
16	52	0.10000	0.02689	0.00235	0.03420	0.00291	-21.35910
16	52	0.20000	0.01771	0.00138	0.03214	0.00742	-44.89074
32	27	0.05000	0.03217	0.00268	0.03901	0.00259	-17.53826
32	27	0.10000	0.02640	0.00342	0.03894	0.00228	-32.21019
32	27	0.20000	0.01705	0.00119	0.03959	0.00234	-56.94200

Table 5 reports the paired seed-wise difference $\Delta\mathrm{RMSE} = \mathrm{RMSE}_{\mathrm{AE}} - \mathrm{RMSE}_{\mathrm{CLP}}$ with Student-t half-widths (df = 2). At steps 16 and 32, where training support is smallest, the mean paired gap is positive at every ρ and CLP-CSGM Ridge wins all three seeds; at step 8, where the development pool is largest, AE $[\boldsymbol{u}, \boldsymbol{b}]$ wins the two smallest ρ cells, matching the mean-RMSE ordering in Table 4. Several of these three-seed intervals include zero. Appendix B.9 repeats the same pairwise comparison on ten seeds: none of those intervals include zero. The CLP-CSGM advantage remains in the few-window cells (steps 16 and 32); AE remains better at step 8 for $\rho = 0.05$ and $\rho = 0.10$. Appendix B.10 repeats the three-seed grid with one extra direct AE that sees an explicit window mask. The mask does not remove the CLP-CSGM advantage at steps 16 and 32. At step 8 the masked AE remains better at $\rho = 0.05$ and $\rho = 0.10$, and the mask does not uniformly lower the direct AE error.

Appendix B.11 redraws the sparse coordinates five times while the model seeds stay at $\{7, 23, 41\}$. The published cell winner is unchanged in all nine cells. At steps 16 and 32, CLP-CSGM Ridge has lower RMSE than AE $[\boldsymbol{u}, \boldsymbol{b}]$ in every draw. At step 8, AE remains better at $\rho = 0.05$ in every draw and at $\rho = 0.10$ on the majority of draws; CLP-CSGM remains better at $\rho = 0.20$ on the majority of draws. The advantage where training support is smallest therefore holds for every tested placement. Step 8, where that support is largest, remains the less stable regime.

Appendix B.12 compares those same nine cells with ordinary least squares and ridge regression from $[\boldsymbol{u}, \boldsymbol{b}]$, and with linear interpolation of the sparse measurements along depth. At the published noise standard deviation 0.01, interpolation has the lowest full-window RMSE in all nine cells and in all 27 seed comparisons. CLP-CSGM Ridge is lower than both direct linear maps at step 16, $\rho = 0.20$, and at step 32 for every ρ. The held-out V_c curve has standard deviation 0.035. Raising only the noise standard deviation to 0.05, the same order as that variation, makes CLP-CSGM Ridge lower in every seed at steps 16 and 32 and at step 8 for $\rho = 0.20$. The penalty weight selected on validation in those cells stays inside the published grid. At 0.10 and 0.20 the CLP-CSGM error remains lower in all nine cells. Those two levels are a sensitivity check: at 0.20 the noise is several times the variation of the curve. The interpolation advantage is tied to the noise level. It holds at the published standard deviation 0.01, where the measurements are a slightly perturbed copy of the curve. Raising that standard deviation to 0.05 removes the advantage in the few-window cells. The interpolation error grows with the noise because the interpolated curve follows the measurements.

The CLP-CSGM error moves less and stays near the window predicted from the dense logs alone. At step 8 for $\rho = 0.05$ and $\rho = 0.10$, interpolation still has the lower mean at noise 0.05. The dense-log starting window at that step already has higher error than the interpolation, and four or six measured depths do not close the gap. The decoder can still represent the curve. These runs are not merged into the submitted tables.

Taken with the ten-seed panel, the two checks separate the randomness behind the three-seed intervals: more seeds, each with its own coordinate draw, and more coordinate draws at fixed model seeds. Both leave the submitted three-seed tables unchanged. Overlapping windows are not counted as extra independent samples.

Table 5: Paired seed-level $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on the cross-well V_c grid ($n = 3$ seeds). SE is the seed-level standard error; the 95% half-width uses Student-t with df = 2. Positive mean ΔRMSE favors CLP-CSGM Ridge. Wins count seeds with strictly lower RMSE for each method.

step	ρ	mean ΔRMSE	SE	t_2 95% half-width	CLP wins	AE wins
8	0.05	-0.00916	0.00350	0.01507	0	3
8	0.10	-0.00451	0.00231	0.00992	1	2
8	0.20	0.00334	0.00200	0.00860	2	1
16	0.05	0.00368	0.00123	0.00531	3	0
16	0.10	0.00730	0.00187	0.00803	3	0
16	0.20	0.01443	0.00362	0.01556	3	0
32	0.05	0.00684	0.00120	0.00515	3	0
32	0.10	0.01254	0.00216	0.00930	3	0
32	0.20	0.02254	0.00202	0.00869	3	0

Figure 6 places the pairwise result within the full benchmark. The mean RMSE of direct MLP and PCA increases sharply as training support decreases, whereas AE has the lowest unweighted mean RMSE among the direct baselines. Table 4 therefore focuses on the most competitive direct comparator in this grid.

Figure 7 shows a representative held-out F03-4 profile for step 16, seed 7, and $\rho = 0.20$. With sonic, bulk density, acoustic impedance, and V_p as dense inputs, CLP-CSGM Ridge follows the broad target variations more closely than AE $[\boldsymbol{u}, \boldsymbol{b}]$ in this example, while direct MLP and PCA are visibly more oscillatory. The profile is consistent with the aggregate RMSE results for this cell; uniform window-level comparisons remain based on the aggregate metrics.

The paired step-16 checks in Appendix B.7 (Table 14) show that including gamma ray reduces absolute error and that the embargo changes training support, but neither modification reverses the ordering of CLP-CSGM Ridge against AE $[\boldsymbol{u}, \boldsymbol{b}]$ and measurement-only CSGM at the tested ratios.

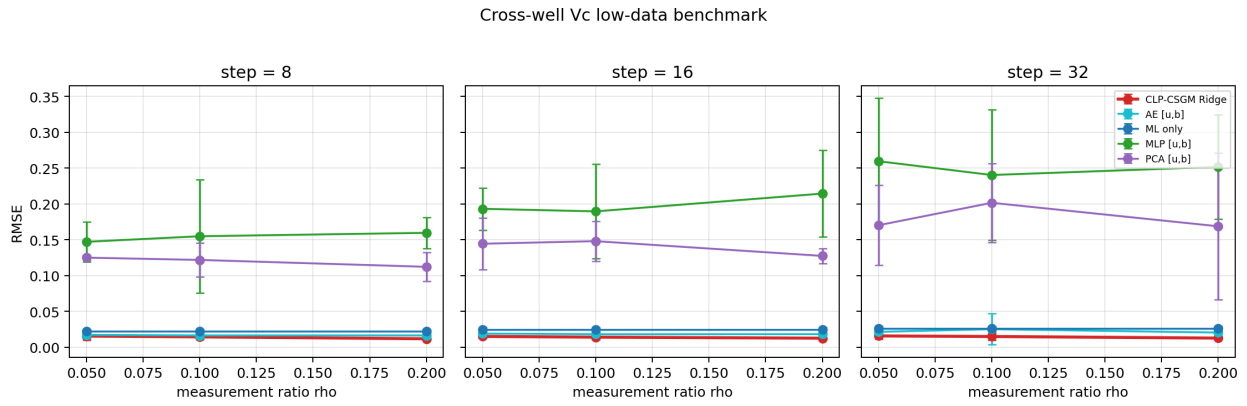


Figure 6: Cross-well V_c RMSE for the main methods. CLP-CSGM Ridge has the lowest mean RMSE in the medium-to-severe low-data cells, while the mean RMSE of direct MLP and PCA increases as training windows become scarce. Error bars are Student-t 95% half-widths across seeds (df = 2).

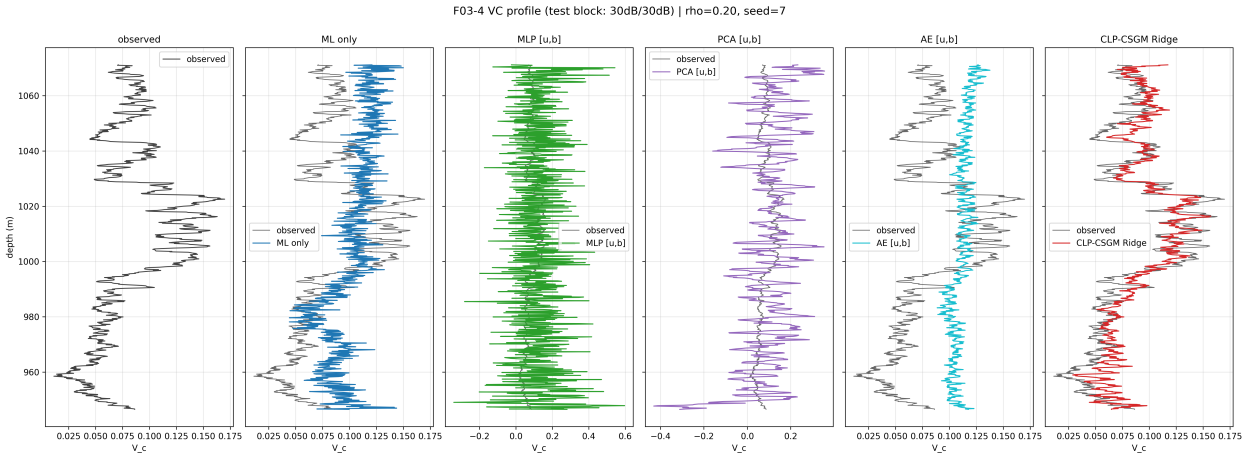


Figure 7: Held-out F03-4 V_c reconstruction at step 16, seed 7, and $\rho = 0.20$. The target is the processed Larionov-derived curve, and the sparse observations are coordinate subsamples of that curve.

6.2 Single-well F03 GR-only results

The F03-4 benchmark uses gamma ray as the only dense log. Table 6 shows that CLP-CSGM Ridge achieves mean RMSE 0.01390 across the tested grid, compared with 0.01434 for AE $[\boldsymbol{u}, \boldsymbol{b}]$. CLP-CSGM Ridge has lower mean RMSE at $\rho \in \{0.20, 0.30\}$, the methods are nearly tied at $\rho = 0.50$, and AE has lower mean RMSE at $\rho \in \{0.40, 0.60\}$. In this benchmark, the largest advantage of explicit latent refinement occurs at the sparsest ratios. Direct AE fusion becomes competitive as more entries of $\boldsymbol{b}$ are available.

Figure 8 displays the complete comparison with seed-level Student-t 95% half-widths (df = 2).

Table 6: Single-well F03-4 GR-only porosity benchmark: mean RMSE by measurement ratio ρ. The prior-only and measurement-only rows are structural reference variants.

method	$\rho = 0.20$	$\rho = 0.30$	$\rho = 0.40$	$\rho = 0.50$	$\rho = 0.60$
CLP-CSGM Ridge	0.01596	0.01475	0.01377	0.01281	0.01220
AE [u,b]	0.01850	0.01554	0.01304	0.01286	0.01178
Prior-only Ridge	0.01958	0.01958	0.01958	0.01958	0.01958
Measurement-only CSGM	0.02519	0.02285	0.02033	0.01622	0.01494
ML only	0.02359	0.02359	0.02359	0.02359	0.02359
PCA [u,b]	0.02149	0.02187	0.02261	0.02409	0.02426
MLP [u,b]	0.03022	0.02997	0.03712	0.04199	0.04627

Table 7 reports the paired $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on the same three seeds: the mean gap is positive at $\rho = 0.20$ and $\rho = 0.30$ (CLP wins all seeds), near zero at $\rho = 0.50$, and negative at $\rho \in \{0.40, 0.60\}$ (AE wins all seeds). With $n = 3$, several paired intervals include zero. The F03 ordering therefore changes with how many depths are measured, and the three-seed intervals are not a large-sample significance claim. RMSE variability, MAE, and relative ℓ_2 error are reported in Appendix B.2 (Table 13) and Appendix B.3 (Figs. 11–12); these metrics retain the same regime-dependent interpretation.

Table 7: Paired seed-level $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on the F03-4 GR-only porosity benchmark ($n = 3$ seeds). SE is the seed-level standard error; the 95% half-width uses Student-t with df = 2. Positive mean ΔRMSE favors CLP-CSGM Ridge.

ρ	mean ΔRMSE	SE	t_2 95% half-width	CLP wins	AE wins
0.20	0.00254	0.00042	0.00180	3	0
0.30	0.00079	0.00032	0.00139	3	0
0.40	-0.00073	0.00035	0.00151	0	3
0.50	0.00006	0.00033	0.00144	2	1
0.60	-0.00041	0.00024	0.00103	0	3

The stress tests in Appendix C (Table 22) extend the sparse grid to $\rho \in \{0.05, 0.10, 0.20\}$. CLP-CSGM Ridge retains lower mean RMSE than AE under increased noise and a smaller training pool, while the ordering becomes less stable with a coarse sliding step and fewer test windows.

Figure 9 shows the depth-profile reconstruction at $\rho = 0.20$, the main-grid ratio at which CLP-CSGM Ridge has its largest mean RMSE advantage over the direct AE baseline.

Because the profile combines predictions from overlapping windows, it provides a qualitative assessment of the overall trend. Pointwise recovery and validation against independent core-porosity measurements fall outside this visualization. At $\rho = 0.20$, CLP-CSGM Ridge follows the broad porosity trend more closely than AE $[\boldsymbol{u}, \boldsymbol{b}]$, although it smooths some local variations under the

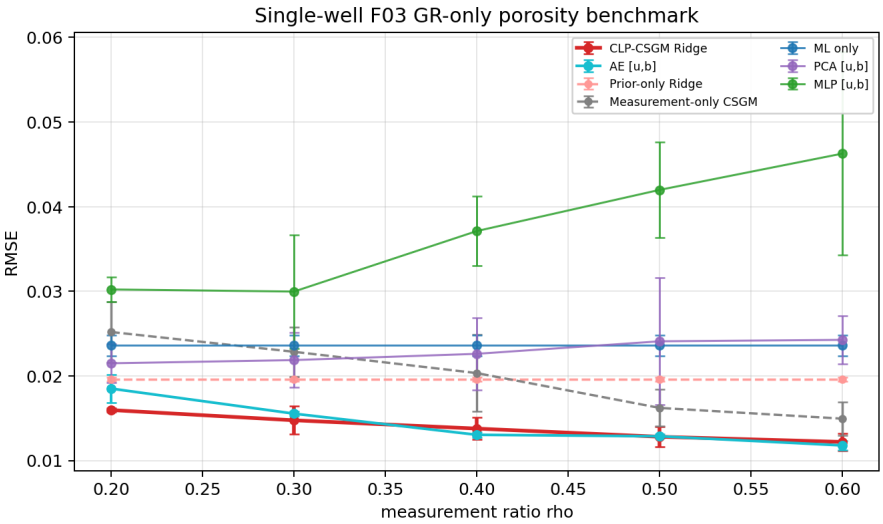


Figure 8: F03-4 GR-only porosity RMSE by measurement ratio. CLP-CSGM Ridge has the lowest mean RMSE at the sparsest ratios, while AE $[\boldsymbol{u}, \boldsymbol{b}]$ becomes competitive as more target observations are available. Error bars are Student-t 95% half-widths across seeds (df $= 2$).

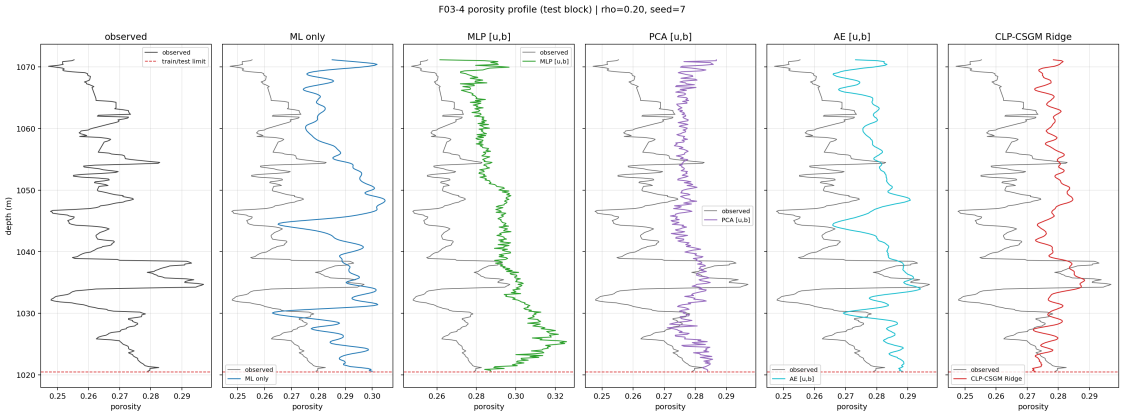


Figure 9: F03-4 depth-profile reconstruction at $\rho = 0.20$. The target is the processed density-derived porosity curve, and the sparse observations are coordinate subsamples of that curve. CLP-CSGM Ridge follows the broad porosity trend while smoothing some local variations.

weak GR-only context. The method uses $m = 13$ observations per length-64 window and achieves a mean RMSE of approximately 0.016. This profile-level behavior is consistent with the lower mean RMSE reported in Table 6 at $\rho = 0.20$. The individual profile is illustrative; the aggregate results support the conclusion that the benefit of explicit latent refinement is concentrated at the sparse end of the tested grid under weak GR-only context. Additional window-level and parity audits are provided in Appendix B.5.

The limited-sample Lapa permeability case in Appendix D provides supplementary evidence. The prior-guided CLP-CSGM variants have lower mean RMSE than the direct baselines that use

Table 8: Cross-well V_c ablation results. Gains are relative RMSE reductions of CLP-CSGM Ridge with respect to the corresponding structural ablation; negative gain indicates that measurement-only CSGM has lower mean RMSE.

Step	ρ	CLP-CSGM	Prior-only	Meas.-only	AE $[u, b]$	Gain vs prior (%)	Gain vs meas. (%)
8	0.0500	0.0448	0.0564	0.0368	0.0357	20.4913	-21.7274
8	0.1000	0.0365	0.0564	0.0297	0.0320	35.3433	-22.7163
8	0.2000	0.0227	0.0564	0.0251	0.0260	59.8016	9.5581
16	0.0500	0.0334	0.0426	0.0362	0.0371	21.5310	7.6727
16	0.1000	0.0269	0.0426	0.0295	0.0342	36.8933	8.9585
16	0.2000	0.0177	0.0426	0.0213	0.0321	58.4391	16.7449
32	0.0500	0.0322	0.0432	0.0341	0.0390	25.5774	5.6452
32	0.1000	0.0264	0.0432	0.0271	0.0389	38.9248	2.4363
32	0.2000	0.0170	0.0432	0.0219	0.0396	60.5584	22.0073

$[\boldsymbol{u}, \boldsymbol{b}]$ across the tested grid. Their comparison with measurement-only CSGM is ratio-dependent.

Appendix E scores laboratory porosity on held-out coordinates of one cored interval. CLP-CSGM is not superior on the mean: the direct AE has the lower RMSE at $\rho = 0.25$ and $\rho = 0.50$ (0.0537 and 0.0523, against 0.0602 and 0.0580 for CLP-CSGM Ridge). Laboratory measurements still correct the log-only anchor, which CLP-CSGM Ridge beats in every split. That correction is enough on the two shallower blocks, where the test logs remain close to the support that fits the anchor and CLP-CSGM Ridge has the lower block-mean RMSE. It does not close the gap on the deepest block, where the logs shift and the anchor starts far from the target windows. The block RMSE, the log summaries, and the split counts are in Appendix E.

6.3 Ablation analysis

The structural ablations drop one term at a time. Prior-only returns $G(h(\boldsymbol{u}))$, the window decoded from the dense-log starting point, with no further adjustment. Measurement-only sets $\lambda = 0$ and searches the decoder range using only the sparse measurements [Bora et al., 2017]. The comparison shows what each term contributes (Remark 1).

In the cross-well benchmark, CLP-CSGM Ridge has lower mean RMSE than prior-only in all nine cells, with reductions from 20.5% to 60.6% (Table 8). It also has lower mean RMSE than measurement-only CSGM in seven cells. The exceptions are step 8 with $\rho \in \{0.05, 0.10\}$, where measurement-only CSGM has lower mean RMSE and the Ridge anchor is not beneficial. At steps 16 and 32, the full model has lower mean RMSE than both ablations at every tested ratio.

In the F03-4 GR-only benchmark, CLP-CSGM Ridge has lower mean RMSE than both ablations

Table 9: Single-well F03-4 GR-only ablation results. Gains are relative RMSE reductions of CLP-CSGM Ridge with respect to the corresponding structural ablation.

ρ	CLP-CSGM	Prior-only	Meas.-only	AE $[u, b]$	Gain vs prior (%)	Gain vs meas. (%)
0.2000	0.0160	0.0196	0.0252	0.0185	18.4646	36.6254
0.3000	0.0148	0.0196	0.0228	0.0155	24.6433	35.4375
0.4000	0.0138	0.0196	0.0203	0.0130	29.6514	32.2641
0.5000	0.0128	0.0196	0.0162	0.0129	34.5823	21.0300
0.6000	0.0122	0.0196	0.0149	0.0118	37.6999	18.3817

Table 10: Cross-well V_c latent-prior sensitivity. Positive signed Ridge gain favors Ridge, whereas negative values favor MLP.

Step	ρ	CLP-CSGM MLP	CLP-CSGM Ridge	Ridge gain (%)
8	0.0500	0.0183	0.0448	-144.8574
8	0.1000	0.0171	0.0365	-112.7948
8	0.2000	0.0138	0.0227	-64.3792
16	0.0500	0.0183	0.0334	-82.4082
16	0.1000	0.0175	0.0269	-53.3581
16	0.2000	0.0141	0.0177	-25.8526
32	0.0500	0.0221	0.0322	-45.4853
32	0.1000	0.0212	0.0264	-24.5143
32	0.2000	0.0158	0.0170	-7.7698

at every tested ratio (Table 9). Its reduction relative to prior-only increases from approximately 18.5% at $\rho = 0.20$ to 37.7% at $\rho = 0.60$, while its largest gains over measurement-only occur at the sparsest ratios. Both terms contribute. The cross-well exceptions show that the ridge starting point is not helpful in every cell.

6.4 Latent-prior model sensitivity

Tables 10 and 11 compare Ridge and MLP implementations of h while keeping the remaining pipeline fixed. This sensitivity analysis is limited to these two prior classes. Signed Ridge gain is defined as $100(\mathrm{RMSE}_{\mathrm{MLP}} - \mathrm{RMSE}_{\mathrm{Ridge}})/\mathrm{RMSE}_{\mathrm{MLP}}$; positive values favor Ridge.

In the cross-well benchmark, MLP has lower mean RMSE in all nine cells. The relative difference is largest at sparse measurement ratios and decreases as ρ increases; at step 16, for example, signed Ridge gain changes from approximately -82% at $\rho = 0.05$ to -26% at $\rho = 0.20$. Ridge remains the headline comparator for the reason in Section 7: a fixed linear reference, not a claim that it dominates MLP on this grid.

In the F03-4 GR-only benchmark, Ridge has lower mean RMSE at every tested ratio, but the relative difference decreases from approximately 14.3% at $\rho = 0.20$ to 1.7% at $\rho = 0.60$.

Table 11: Single-well F03-4 GR-only latent-prior sensitivity. Signed Ridge gain follows the definition used in Table 10.

ρ	CLP-CSGM MLP	CLP-CSGM Ridge	Ridge gain (%)
0.2000	0.0186	0.0160	14.2682
0.3000	0.0167	0.0148	11.4868
0.4000	0.0152	0.0138	9.2929
0.5000	0.0132	0.0128	2.6767
0.6000	0.0124	0.0122	1.6615

Which map is better depends on the dataset and on how many depths are measured, and the difference shrinks as more target values enter $\boldsymbol{b}$. Ridge stays the reported reference because it is a fixed linear map, not because it wins every cell. The rankings support choosing that map on validation data for the intended test, rather than after seeing the held-out result.

Supplementary diagnostics. Appendix B.6 audits validation, selected λ, and per-window errors; Appendix B.4 shows realized sparse-measurement windows; Appendix B.1 assesses empirical decoder-range separation; and Appendix B.13 reports inference cost. These analyses serve interpretive and reproducibility purposes and are separate from the benchmark comparisons.

7 Interpretation, scope, and limitations

CLP-CSGM uses three pieces that act together. The decoder range $\mathrm{Im}(G)$ is the set of target windows the decoder can produce. The starting code $\boldsymbol{z}_0 = h(\boldsymbol{u})$ is the point the dense logs propose before any measurement is enforced. The measurement term $\|M_\Omega G(\boldsymbol{z}) - \boldsymbol{b}\|_2^2$ pulls the decoded window onto the sparse observations. The decoder range limits which windows can appear. The starting code carries the dense logs into that search, and the measurement term uses the sparse target values. Unlike direct $[\boldsymbol{u}, \boldsymbol{b}]$ predictors, CLP-CSGM retains $\boldsymbol{b}$ as an explicit measurement during inference. It is therefore most relevant when dense logs are informative but incomplete and sparse target entries should be honored directly. Log-completion models fill a missing interval of a wireline curve, geomodel translation fills a property image from another image, and direct regression maps logs to porosity or shale volume [Koeshidayatullah et al., 2026, Al-Fakih et al., 2026, 2025, 2023]. CLP-CSGM is the relevant formulation when those dense logs are already available and a few observations of the target property must be honored by the reconstructed window.

The depth profiles in Figs. 7 and 9 illustrate how the balance of these information sources affects

reconstruction detail. The multi-channel cross-well context supports closer tracking of the held-out V_c profile, whereas the deliberately weak GR-only context produces a smoother F03-4 porosity reconstruction. The attainable detail is governed jointly by the information in $\boldsymbol{u}$, the adequacy of $\mathrm{Im}(G)$, and the number and placement of observations in $\boldsymbol{b}$.

The ablations in Section 6.3 drop one piece at a time. Using only the starting point, or only the measurements, raises the error in most cells, and which piece matters more changes with the number of training windows. The full Ridge model has lower mean RMSE than prior-only throughout the cross-well grid and lower mean RMSE than measurement-only CSGM in seven of nine cells. In the two step-8 exceptions, measurement-only inversion has lower mean RMSE, so the ridge starting point is not helpful in those cells. On the F03-4 grid, where gamma ray is the only dense log, the full model has lower mean RMSE than both ablations at every tested ratio. The starting point keeps the search from drifting, and the measurement term is still free to correct it.

The sensitivity results in Section 6.4 show that no single architecture for h is uniformly preferable. MLP has lower mean RMSE in all nine multi-channel cross-well cells, whereas Ridge has lower mean RMSE at every tested ratio under the single-channel F03-4 setting. In both cases, the difference decreases as ρ increases, which is consistent with less dependence on the dense-log starting point as more target observations become available. These comparisons distinguish the CLP-CSGM framework from its reference implementation. Ridge remains the primary reported configuration because it is a fixed linear map: the main tables then use one h in every cell, and h is not chosen after seeing the test ranking. The MLP results have the same experimental status. They show that a more flexible map improves the cross-well grid and that the preferred class changes with the test. The class of h, the latent dimension, and the regularization should be selected on validation data for the intended measurements, not by adopting whichever variant wins on the held-out test set. The experiments establish this sensitivity but do not identify which architectural property causes the opposite rankings.

The empirical scope is limited by the construction of the benchmark targets. In the cross-well experiment, V_c is a processed Larionov-derived curve. The gamma-ray index between the clean and shale endpoints is linear in gamma ray. The Larionov map from that index to clay volume is not a straight line. If gamma ray has been logged, clay volume is obtained from that relation, and CLP-CSGM is not a substitute for it. Gamma ray is left out of the main input because the clay-volume

curve was computed from it. Without that log the Larionov relation cannot be evaluated. The task is then to reconstruct the processed curve from the other dense logs while matching a few known values of the curve. Recovery of independently acquired clay-volume measurements is outside this scope. Putting gamma ray back, in Appendix B.7, lowers the absolute error and does not change the ordering of CLP-CSGM Ridge, the direct AE, and measurement-only CSGM at step 16. That run checks the channel. It is not the log suite used for the main comparison.

The F03-4 porosity curve is density-derived, and its sparse values are taken from that same curve, not from core or laboratory plugs. Density porosity is linear in bulk density, and a standard sonic porosity is linear in slowness. If those logs have been recorded, the processed porosity is already determined. Sonic travel time is left out because it follows this curve almost one to one, and the density curve behind the porosity was itself predicted from sonic and gamma ray. The test uses gamma ray as the only dense log. Its errors do not transfer to a porosity interpretation that also uses density, neutron, and sonic. The two F3 benchmarks therefore remove the log the conventional formula needs and compare methods that must reconstruct the processed curve from the remaining logs while matching a few known values. They are not a contest against Larionov, density porosity, or sonic porosity on a fully logged well, and they are not a test against independent core or laboratory measurements.

The Lapa permeability case [Alvarez et al., 2025] extends the evaluation to another lithology and target property, but its small sample size limits generalization. The laboratory-porosity supplement in Appendix E uses $\mathrm{Phi}_{\mathrm{lab}}$ rather than a curve computed from the wireline suite, and it scores only coordinates that were not supplied as measurements. The direct AE has the lower pooled mean RMSE. Laboratory coordinates still correct the log-only anchor in every split, and that correction is enough where the test logs remain close to the blocks that fit $h(\boldsymbol{u})$. On the deepest block the logs move outside that support, the ridge anchor starts far from the target windows, and a few measurements improve on the anchor without reaching the direct AE. The measurement term still lowers the error relative to the log-only anchor in that block. The supplement therefore does not extend the F3 low-support advantage to a mean win on this laboratory target. Each test fold still has many overlapping training windows, so the direct map is in the regime where it was expected to be competitive. Block RMSE and the log summaries are reported in the appendix. Broader validation requires larger multi-well datasets, additional log suites, and facies-conditioned settings.

The reported metrics are not stratified by lithofacies, lithological texture, or formal stratigraphic interval because the available benchmark tables do not provide independent geological labels at each depth sample. Consequently, RMSE, MAE, and relative ℓ_2 error aggregate heterogeneous windows. The depth profiles provide qualitative audits but no geological performance breakdowns. Facies-resolved reliability would require independently interpreted or core-calibrated interval labels, particularly because the adequacy of the decoder range and the accuracy of the latent prior may vary across geological regimes.

The recovery interpretation is also conditional. Proposition 3 relates reconstruction error to decoder mismatch, latent-prior mismatch, measurement noise, optimization quality, and local separability, but does not prove that M_Ω satisfies S-REC for the trained decoder. The diagnostics in Appendix B.1 evaluate sampled decoder outputs and should not be interpreted as an operator-level guarantee.

CLP-CSGM incurs iterative inference cost, whereas direct baselines produce predictions through a forward pass. The runtime results in Appendix B.13 show that this additional cost is most justified when explicit measurement-consistent refinement produces a meaningful accuracy gain. Faster latent solvers, warm starts, adaptive stopping, or amortized refinement are natural directions for larger-scale deployment.

8 Conclusion

We introduced CLP-CSGM. A decoder trained on windows of the target curve supplies the windows the reconstruction may use. The dense logs set the starting point of the search. At inference the decoded window must match the sparse measurements. The MAP/Tikhonov reading and the local error bound separate measurement noise, decoder error, a poor starting point, and whether nearby windows can be told apart from the measurements.

Under the paired F3 protocols, CLP-CSGM Ridge has lower mean RMSE than AE $[\boldsymbol{u}, \boldsymbol{b}]$, the most competitive direct baseline in this cross-well benchmark, in seven of nine cross-well cells. This is a conditional advantage where training support is smallest (steps 16 and 32), not overall superiority: AE $[\boldsymbol{u}, \boldsymbol{b}]$ has lower mean RMSE at step 8 for $\rho = 0.05$ and $\rho = 0.10$. The unweighted mean of the nine cell-level mean RMSE values decreases from 0.03496 to 0.02862, a reduction of

approximately 18.1%. In the F03-4 GR-only benchmark, CLP-CSGM Ridge has the lowest mean RMSE at $\rho \in \{0.20, 0.30\}$, while AE $[\boldsymbol{u}, \boldsymbol{b}]$ has lower mean RMSE at $\rho \in \{0.40, 0.60\}$. That benchmark assimilates the processed porosity curve with gamma ray as the only dense input; the depth profile at $\rho = 0.20$ follows the broad trend of this curve. It is a controlled stress test under intentionally incomplete context and is not evidence of general porosity reconstruction performance. The full Ridge model has lower mean RMSE than prior-only in every cross-well cell and than measurement-only CSGM in seven of nine cells. It also has lower mean RMSE than both ablations throughout the F03-4 grid. The opposite Ridge–MLP rankings show that the better map from dense logs to the starting point depends on the test.

On the tests reported here, CLP-CSGM complements direct regression because the sparse values stay measurements that the reconstructed window must match. The F3 evidence is reconstruction of processed target logs: cross-well V_c is Larionov-derived from gamma ray, F03-4 porosity is density-derived, and sparse observations are coordinate subsamples of these same curves rather than independently acquired core or laboratory measurements. A separate single-well supplement on laboratory porosity scores held-out laboratory coordinates. CLP-CSGM is not superior on the mean RMSE: the direct AE is lower because the deepest block shifts the logs away from the logs used to fit the starting-point map. Laboratory measurements correct that starting point in every split, and the correction is enough on the two shallower blocks, where the test logs remain close to those training logs and CLP-CSGM Ridge keeps the lower block-mean RMSE. That supplement is one cored interval. The cross-well comparison is a controlled single-holdout benchmark on F03-4 and is not evidence of operational cross-well or multi-field transferability. Broader assessment will require larger multi-well datasets, independently calibrated targets, and geological labels. It will also require uncertainty estimates and faster latent-refinement strategies.

Author Contributions

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI's ChatGPT to assist with language refinement, improve the clarity and organization of the manuscript, reduce its length to meet the journal's word-count requirements, and assist in the creation of explanatory schematic figures illustrating concepts, processes, and experimental workflows. The tool was not used to generate research data, perform the reported numerical experiments, fabricate or alter quantitative results, or determine the scientific conclusions. After using this tool, the authors reviewed, edited, and validated the text and figures as needed and take full responsibility for the content of the published article.

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A Supplementary Theoretical Details

This section collects the proofs omitted from the main text, including the detailed conditional local-recovery argument. That argument separates four contributions to the error: how far the true window is from anything the decoder can produce, how far the dense-log starting code is from a good code, measurement noise, and whether the measurements can tell nearby windows apart.

A.1 MAP and Tikhonov interpretation

Proof of Proposition 1. The MAP estimate is obtained by maximizing the posterior density

$$p(z \mid b, u).$$

By Bayes' rule,

$$p(z \mid b, u) = \frac{p(b \mid z, u)p(z \mid u)}{p(b \mid u)}.$$

Using the conditional independence assumption $p(b \mid z, u) = p(b \mid z)$, and noting that $p(b \mid u)$ does not depend on z, we obtain

$$p(z \mid b, u) \propto p(b \mid z)p(z \mid u).$$

Therefore,

$$\hat{z}_{\mathrm{MAP}} \in \arg\min_z \left[-\log p(b \mid z) - \log p(z \mid u) \right].$$

Since

$$b \mid z \sim \mathcal{N}(M_\Omega G(z), \sigma_\eta^2 I_m),$$

we have, up to constants independent of z,

$$-\log p(b \mid z) = \frac{1}{2\sigma_\eta^2} \|M_\Omega G(z) - b\|_2^2 + \text{const}.$$

Similarly, since

$$z \mid u \sim \mathcal{N}(z_0(u), \sigma_z^2 I_k),$$

we have, up to constants independent of z,

$$-\log p(z \mid u) = \frac{1}{2\sigma_z^2}\|z - z_0(u)\|_2^2 + \text{const.}$$

Hence the MAP estimate minimizes

$$\frac{1}{2\sigma_\eta^2}\|M_\Omega G(z) - b\|_2^2 + \frac{1}{2\sigma_z^2}\|z - z_0(u)\|_2^2.$$

Multiplying this objective by the positive constant $2\sigma_\eta^2$ does not change its minimizers. Thus,

$$\hat{z}_{\text{MAP}} \in \arg\min_z \|M_\Omega G(z) - b\|_2^2 + \frac{\sigma_\eta^2}{\sigma_z^2}\|z - z_0(u)\|_2^2.$$

Therefore the CLP-CSGM objective is recovered with

$$\lambda = \frac{\sigma_\eta^2}{\sigma_z^2}.$$

The second term is a Tikhonov regularization term centered at the log-conditioned latent anchor $z_0(u)$, not at the origin. □

A.2 Existence of a minimizer

Proof of Proposition 2. Since G is continuous and M_Ω is linear, the map

$$z \mapsto \|M_\Omega G(z) - b\|_2^2$$

is continuous. The quadratic term

$$z \mapsto \lambda\|z - z_0\|_2^2$$

is also continuous. Hence

$$J(z) = \|M_\Omega G(z) - b\|_2^2 + \lambda\|z - z_0\|_2^2$$

is continuous on $\mathbb{R}^k$.

Moreover, since $\lambda > 0$,

$$J(z) \geq \lambda\|z - z_0\|_2^2.$$

As $\|z\|_2 \to \infty$, we also have $\|z - z_0\|_2 \to \infty$. Therefore,

$$J(z) \to +\infty \qquad \text{as} \qquad \|z\|_2 \to \infty.$$

Thus J is coercive.

Choose the reference level

$$c = J(z_0),$$

and define the sublevel set

$$S_c = \{z \in \mathbb{R}^k : J(z) \leq c\}.$$

The set S_c is nonempty because $z_0 \in S_c$. It is closed since

$$S_c = J^{-1}((-\infty, c]),$$

J is continuous, and $(-\infty, c]$ is closed in $\mathbb{R}$.

Since J is coercive, there exists $R > 0$ such that

$$\|z\|_2 > R \quad \Longrightarrow \quad J(z) > c.$$

Therefore no point with $\|z\|_2 > R$ belongs to S_c, and hence

$$S_c \subset \overline{B}_R = \{z \in \mathbb{R}^k : \|z\|_2 \leq R\}.$$

Thus S_c is bounded. Since S_c is closed and bounded in the finite-dimensional space $\mathbb{R}^k$, it is compact.

By the Weierstrass theorem, the continuous function J attains its minimum on S_c. Hence there exists $\hat{z} \in S_c$ such that

$$J(\hat{z}) = \min_{z \in S_c} J(z).$$

It remains only to note that this minimum is also global. Indeed, if $z \in S_c$, then

$$J(\hat{z}) \leq J(z)$$

by the definition of $\hat{z}$. If $z \notin S_c$, then $J(z) > c$, while $\hat{z} \in S_c$ implies

$$J(\hat{z}) \leq c.$$

Therefore,

$$J(\hat{z}) \leq c < J(z)$$

for every $z \notin S_c$. Hence no point outside S_c can have objective value smaller than $J(\hat{z})$. Consequently,

$$J(\hat{z}) = \min_{z \in \mathbb{R}^k} J(z),$$

and J admits at least one global minimizer in $\mathbb{R}^k$. □

A.3 Conditional local recovery interpretation

Proposition 3 gives the conditional local recovery interpretation summarized in the main text. The full proof is followed by the corresponding extension for a numerical optimizer.

Proof of Proposition 3. By global optimality of $\hat{z}$,

$$\mathcal{J}(\hat{z}) \leq \mathcal{J}(z^\dagger).$$

Since $y^\star = G(z^\dagger) + r^\dagger$, we have

$$b = M_\Omega y^\star + \eta = M_\Omega G(z^\dagger) + M_\Omega r^\dagger + \eta = M_\Omega G(z^\dagger) + e^\dagger.$$

Therefore,

$$\mathcal{J}(z^\dagger) = \|M_\Omega G(z^\dagger) - b\|_2^2 + \lambda \|z^\dagger - z_0\|_2^2 = \|e^\dagger\|_2^2 + \lambda \|z^\dagger - z_0\|_2^2 = (\Delta^\dagger)^2.$$

Hence $\mathcal{J}(\hat{z}) \leq (\Delta^\dagger)^2$. Since each term in $\mathcal{J}(\hat{z})$ is nonnegative,

$$\lambda \|\hat{z} - z_0\|_2^2 \leq \mathcal{J}(\hat{z}) \leq (\Delta^\dagger)^2,$$

which yields

$$\|\hat{z} - z_0\|_2 \le \frac{\Delta^\dagger}{\sqrt{\lambda}}.$$

Also, by the definition of $\Delta^\dagger$,

$$\Delta^\dagger \ge \sqrt{\lambda}\|z^\dagger - z_0\|_2,$$

so

$$\|z^\dagger - z_0\|_2 \le \frac{\Delta^\dagger}{\sqrt{\lambda}}.$$

Thus both $\hat{z}$ and $z^\dagger$ belong to $B(z_0, R^\dagger)$.

The local S-REC on $\mathcal{M}_{R^\dagger}$ therefore applies to $G(\hat{z})$ and $G(z^\dagger)$, giving

$$\gamma_R\|G(\hat{z}) - G(z^\dagger)\|_2 - \delta_R \le \|M_\Omega(G(\hat{z}) - G(z^\dagger))\|_2.$$

By the triangle inequality,

$$\|M_\Omega(G(\hat{z}) - G(z^\dagger))\|_2 \le \|M_\Omega G(\hat{z}) - b\|_2 + \|b - M_\Omega G(z^\dagger)\|_2.$$

The first term satisfies

$$\|M_\Omega G(\hat{z}) - b\|_2^2 \le \mathcal{J}(\hat{z}) \le (\Delta^\dagger)^2,$$

so $\|M_\Omega G(\hat{z}) - b\|_2 \le \Delta^\dagger$. The second term is $\|e^\dagger\|_2$. Hence

$$\|M_\Omega(G(\hat{z}) - G(z^\dagger))\|_2 \le \Delta^\dagger + \|e^\dagger\|_2.$$

Combining the last two displays gives

$$\|G(\hat{z}) - G(z^\dagger)\|_2 \le \frac{\Delta^\dagger + \|e^\dagger\|_2 + \delta_R}{\gamma_R}.$$

Finally, since $y^\star = G(z^\dagger) + r^\dagger$, a triangle inequality yields

$$\|G(\hat{z}) - y^\star\|_2 \le \|G(\hat{z}) - G(z^\dagger)\|_2 + \|r^\dagger\|_2 \le \|r^\dagger\|_2 + \frac{\Delta^\dagger + \|e^\dagger\|_2 + \delta_R}{\gamma_R}.$$

□

The same reasoning extends to a numerical solution $\hat{z}_{\mathrm{num}}$ produced by an iterative optimizer such as Adam. If

$$\mathcal{J}(\hat{z}_{\mathrm{num}}) \le \mathcal{J}(z^\dagger) + \tau,$$

then one replaces $(\Delta^\dagger)^2$ by $(\Delta^\dagger)^2 + \tau$ throughout the argument, so the bounds hold up to an additional optimization tolerance.

Coordinate restriction and incoherent secants. The local S-REC assumption above is not automatic for a coordinate-restriction operator. If $A_\Omega = \sqrt{N/m}\, M_\Omega$ denotes the rescaled restriction operator and Ω is sampled uniformly, then for any fixed $v \in \mathbb{R}^N$,

$$\mathbb{E}\|A_\Omega v\|_2^2 = \|v\|_2^2.$$

Thus, random coordinate restriction preserves the energy of a fixed vector in expectation. However, concentration around this expectation requires that the energy of v is not concentrated on a small number of coordinates. A standard sufficient condition is an incoherence bound of the form

$$\|v\|_\infty^2 \le \frac{\mu}{N}\|v\|_2^2.$$

Under such a condition, random coordinate samples concentrate $\|A_\Omega v\|_2^2$ around $\|v\|_2^2$. For a finite set of candidate signals, a union bound extends this concentration to all pairwise secants, provided the number of sampled coordinates is large enough relative to the incoherence level and the logarithm of the number of pairs. For the continuous decoder range, the analogous requirement would need to hold uniformly over the normalized secants

$$\frac{G(z_1) - G(z_2)}{\|G(z_1) - G(z_2)\|_2}, \qquad z_1, z_2 \in B(z_0, R^\dagger).$$

This strong assumption is not verified analytically for the trained decoder. We therefore do not claim that M_Ω satisfies S-REC over the learned decoder range. The local S-REC serves only as a sufficient theoretical lens; Appendix B.1 empirically assesses the practical separability of decoder-generated candidates under coordinate subsampling.

B Supplementary Diagnostics

B.1 Measurement Separability

Table 12 and Fig. 10 provide an empirical check of the separation property that is relevant for generative compressed sensing. For pairs of decoder-generated training windows, we compute

$$r_{ij} = \frac{\sqrt{N/m}\,\|M_\Omega(y_i - y_j)\|_2}{\|y_i - y_j\|_2}.$$

Values close to one indicate that coordinate subsampling preserves pairwise distances after energy normalization. This diagnostic does not prove S-REC over the learned decoder range. It instead examines the corresponding practical question of whether sparse observations distinguish plausible generated windows. The median ratio remains close to one in all tested cells. Lower percentiles are generally stronger in denser settings, despite some non-monotonic variation at intermediate ρ values. The weakest case is the sparsest cross-well setting, where only four entries are observed per window, consistent with the larger ill-posedness expected at very low measurement ratios.

Table 12: Empirical S-REC-style separation diagnostics on decoder-generated windows. Values are percentiles of the energy-normalized ratio $\sqrt{N/m}\|M_\Omega(y_i - y_j)\|_2/\|y_i - y_j\|_2$.

Dataset	ρ	m	P1	P5	P10	Median
Cross-well V_c	0.05	4	0.320	0.483	0.567	0.963
Cross-well V_c	0.10	6	0.291	0.527	0.646	1.004
Cross-well V_c	0.20	13	0.770	0.882	0.930	1.039
F03-4 GR-only	0.20	13	0.641	0.753	0.815	1.000
F03-4 GR-only	0.40	26	0.600	0.767	0.833	0.999
F03-4 GR-only	0.60	38	0.856	0.902	0.927	0.998

B.2 Full F03-4 benchmark table

Table 13 reports the complete single-well F03-4 GR-only porosity benchmark used in Section 6.2. In addition to mean RMSE, it includes seed-level RMSE variability (standard deviation and Student-t 95% half-width with df = 2), MAE, and relative ℓ_2 error for all seven methods at each tested

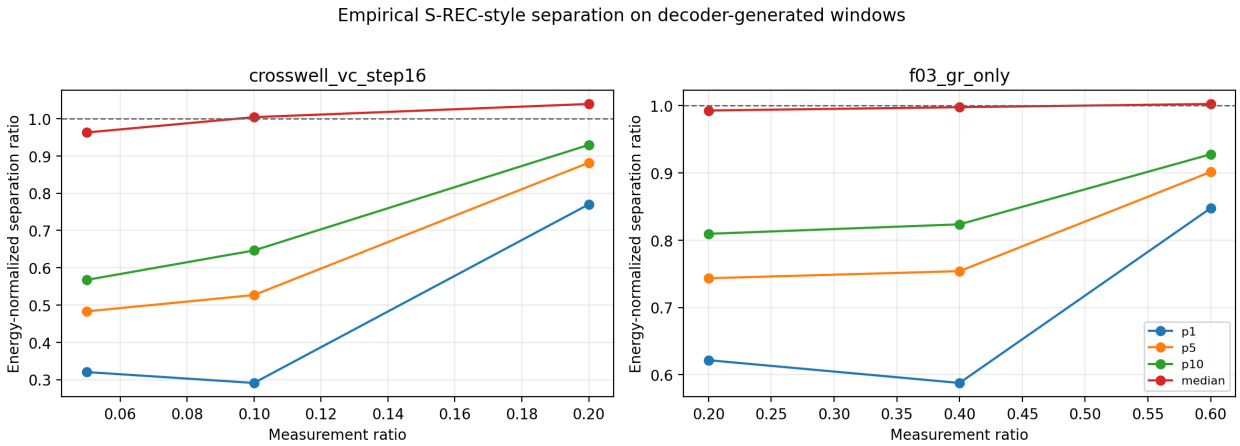


Figure 10: Empirical S-REC-style separation diagnostics. The curves show lower percentiles and medians of the energy-normalized pairwise separation ratio on decoder-generated windows. Higher lower-percentile values indicate stronger separation of plausible generated candidates under coordinate subsampling.

measurement ratio.

B.3 F03-4 Metric Robustness Views

The F03-4 GR-only discussion in Section 6.2 uses RMSE because the protocol selects and reports models on that scale. Figures 11 and 12 evaluate the same interpretation under alternative error summaries. MAE downweights isolated large deviations relative to RMSE. Relative ℓ_2 error indicates whether the ranking mainly reflects a scale effect. Both plots retain the sparse-regime pattern: CLP-CSGM Ridge is lowest or close to lowest at lower measurement ratios, and AE $[u, b]$ becomes more competitive as more entries of b are available. The degradation of direct MLP $[u, b]$ also appears in more than one metric, consistent with sparse measurements acting as high-variance features when the measurement operator is not handled explicitly.

B.4 Illustrative sparse-measurement windows

The main tables and curves in Section 6 pool scalar errors over many test windows. The examples in Figs. 13 and 14 display the target profile, the coordinates where b is observed, and the full-window predictions generated under the paired (seed, ρ) design used in the main experiments, thereby complementing the aggregate comparisons with window-level views.

The horizontal axis indexes coordinates within one length-L window ($L = 64$). The blue curve

Table 13: Full single-well F03-4 GR-only porosity benchmark by measurement ratio, including direct baselines, structural reference variants, and multi-metric summaries across seeds. The column rmse_t95_half is the Student-t 95% half-width for the seed mean (df = 2), not a normal $z = 1.96$ interval.

rho	method	rmse	rmse_std	rmse_t95_half	mae	relative_l2
0.20000	CLP-CSGM Ridge	0.01596	0.00010	0.00024	0.01409	0.06000
0.20000	AE [u,b]	0.01850	0.00067	0.00166	0.01619	0.06951
0.20000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.20000	PCA [u,b]	0.02149	0.00093	0.00230	0.01796	0.08073
0.20000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.20000	Measurement-only CSGM	0.02519	0.00145	0.00360	0.01993	0.09459
0.20000	MLP [u,b]	0.03022	0.00059	0.00146	0.02588	0.11356
0.30000	CLP-CSGM Ridge	0.01475	0.00067	0.00166	0.01294	0.05544
0.30000	AE [u,b]	0.01554	0.00011	0.00027	0.01351	0.05837
0.30000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.30000	PCA [u,b]	0.02187	0.00130	0.00323	0.01799	0.08214
0.30000	Measurement-only CSGM	0.02285	0.00117	0.00290	0.01786	0.08578
0.30000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.30000	MLP [u,b]	0.02997	0.00270	0.00670	0.02555	0.11253
0.40000	AE [u,b]	0.01304	0.00013	0.00033	0.01116	0.04897
0.40000	CLP-CSGM Ridge	0.01377	0.00052	0.00129	0.01197	0.05176
0.40000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.40000	Measurement-only CSGM	0.02033	0.00182	0.00452	0.01560	0.07635
0.40000	PCA [u,b]	0.02261	0.00172	0.00427	0.01839	0.08490
0.40000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.40000	MLP [u,b]	0.03712	0.00166	0.00412	0.03244	0.13932
0.50000	CLP-CSGM Ridge	0.01281	0.00048	0.00119	0.01110	0.04813
0.50000	AE [u,b]	0.01286	0.00010	0.00025	0.01107	0.04832
0.50000	Measurement-only CSGM	0.01622	0.00087	0.00217	0.01281	0.06089
0.50000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.50000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.50000	PCA [u,b]	0.02409	0.00301	0.00748	0.01952	0.09050
0.50000	MLP [u,b]	0.04199	0.00227	0.00564	0.03757	0.15764
0.60000	AE [u,b]	0.01178	0.00023	0.00058	0.01003	0.04425
0.60000	CLP-CSGM Ridge	0.01220	0.00041	0.00102	0.01051	0.04583
0.60000	Measurement-only CSGM	0.01494	0.00081	0.00200	0.01176	0.05611
0.60000	Prior-only Ridge	0.01958	0.00009	0.00022	0.01774	0.07360
0.60000	ML only	0.02359	0.00051	0.00126	0.02086	0.08866
0.60000	PCA [u,b]	0.02426	0.00114	0.00284	0.01976	0.09109
0.60000	MLP [u,b]	0.04627	0.00483	0.01199	0.04076	0.17381

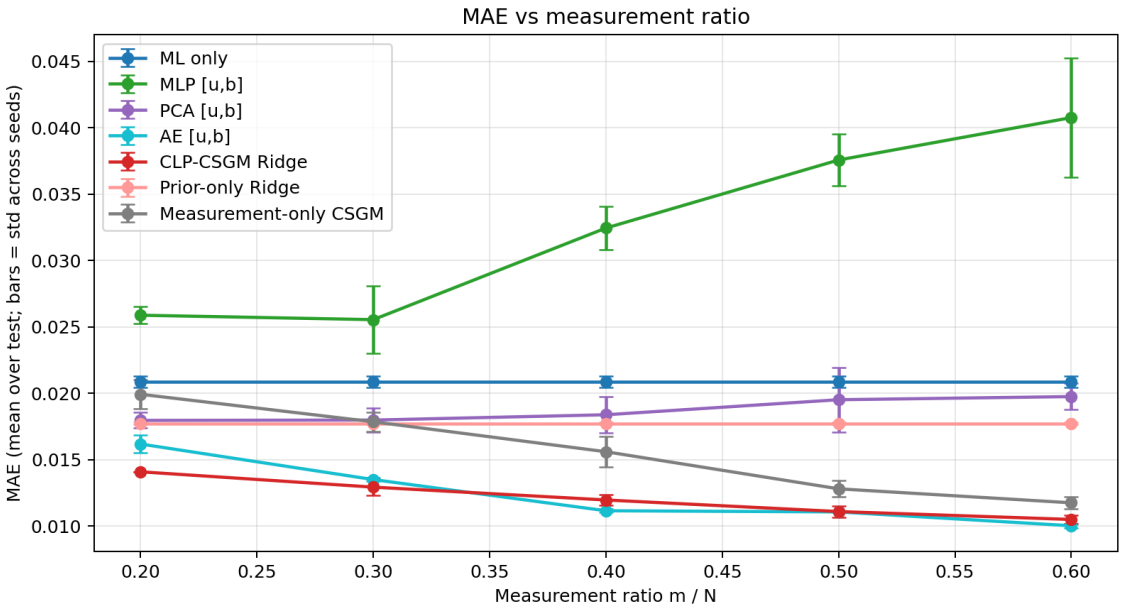


Figure 11: Single-well F03-4 GR-only MAE by measurement ratio. The ranking is consistent with the RMSE view in the main text: CLP-CSGM Ridge has the lowest MAE in the sparse regime and remains competitive as the measurement ratio increases.

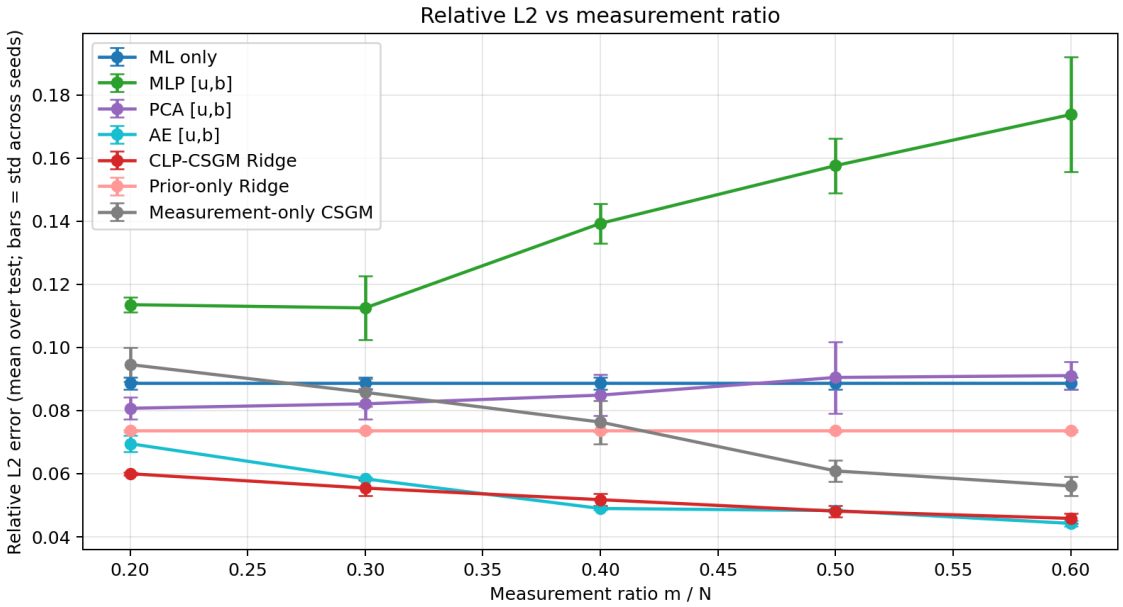


Figure 12: Relative ℓ_2 error on the single-well F03-4 GR-only benchmark. The relative-error view preserves the same conclusion as RMSE and MAE, indicating that the gain is not caused by scale effects alone.

is the target window y. Black markers show the realized noisy measurements b at the sampled coordinates. The red curve is CLP-CSGM Ridge and the green curve is AE $[u, b]$, when shown. These markers are the actual sparse inputs generated under the benchmark noise model.

Each figure fixes one test-window index, one seed, and one ρ. These audit examples show that the markers alone underdetermine the reconstruction and that the methods interpolate between sparse constraints using different inductive biases. Aggregate tables and the main-text diagnostics provide the overall performance comparisons.

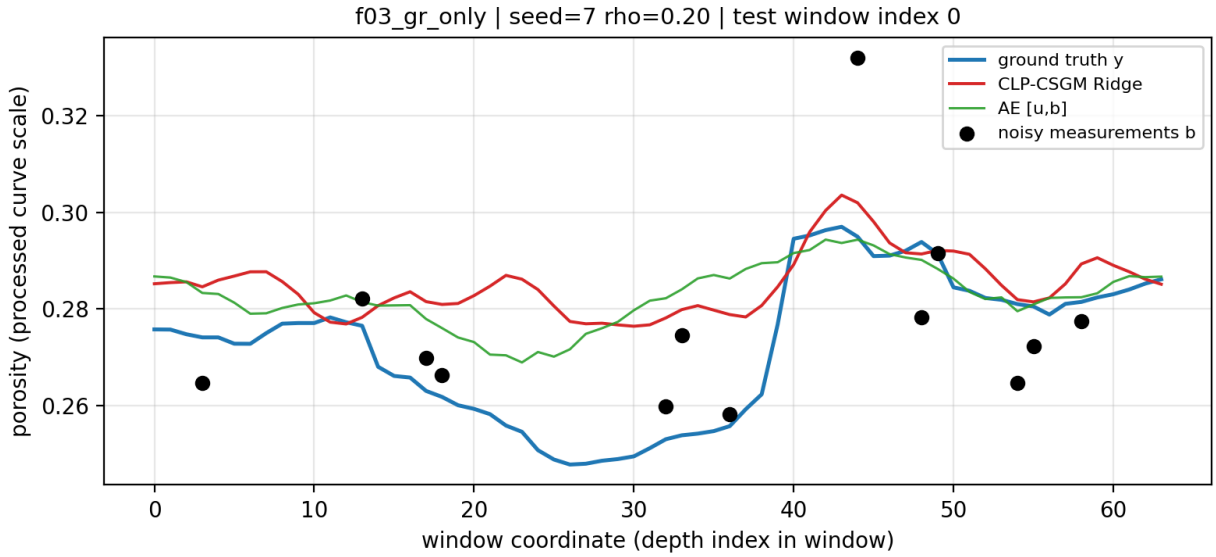


Figure 13: F03-4 GR-only porosity example (test window index 0, seed 7, $\rho = 0.20$): ground truth, sparse noisy measurements b, CLP-CSGM Ridge, and AE $[u, b]$.

B.5 F03-4 Qualitative Reconstruction Views

This subsection complements the depth-profile reconstruction in the main text (Fig. 9). Because the depth profile follows the contiguous test interval, it provides the clearest qualitative view. The figures below present window-level and pointwise audits.

Figure 15 shows representative F03-4 test windows and reveals local shape behavior compressed by scalar metrics. Direct MLP $[u, b]$ can produce oscillatory predictions, whereas PCA, AE, and CLP-CSGM Ridge are smoother. Aggregate metrics determine the window-level rankings. Figure 16 pools pointwise predictions against ground truth over the F03-4 test windows. CLP-CSGM Ridge has the highest displayed pointwise correlation among the compared methods, but the scatter

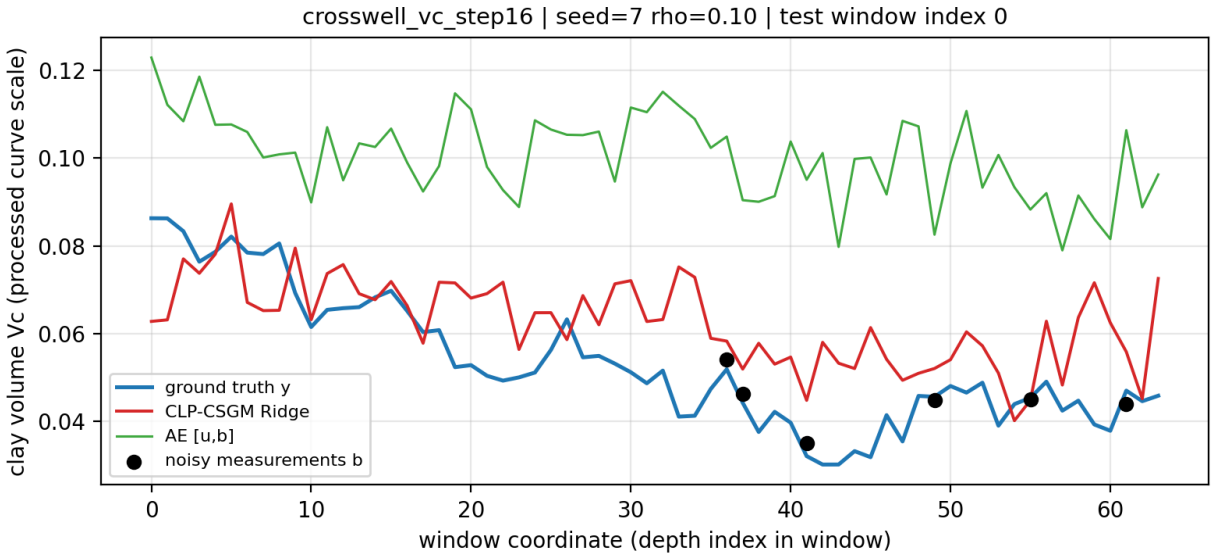


Figure 14: Cross-well V_c example (held-out F03-4 test window index 0, step 16, seed 7, $\rho = 0.10$): ground truth, sparse noisy measurements b, CLP-CSGM Ridge, and AE $[u, b]$.

remains broad. The plot therefore indicates improved relative agreement while retaining substantial pointwise variability.

B.6 Validation and Generalization Diagnostics

Figures 17 and 18 audit quantities used by the experimental protocol: validation RMSE for selecting λ, selected regularization values, per-window test RMSE distributions, and the relation between autoencoder training reconstruction error and final test RMSE. These diagnostics support the reliability and interpretation of the main rankings.

Figure 17 summarizes the cross-well V_c case. The validation and test RMSE values remain of similar magnitude across the low-data grid, with no sign that lower validation error is obtained only by shifting error to the test well. The selected λ values are small but nonzero, consistent with sparse target-log observations providing the main target-well information while the log-conditioned latent prior acts as a stabilizing regularizer. The per-window RMSE boxplots reinforce the ablation result: CLP-CSGM Ridge has a lower median error than the latent-prior-only decoder and a substantially tighter distribution than measurement-only CSGM.

Figure 18 gives the analogous diagnostic view for the F03-4 porosity benchmark with GR-only dense input. The validation–test relation is less diagonal than in the cross-well case, consistent

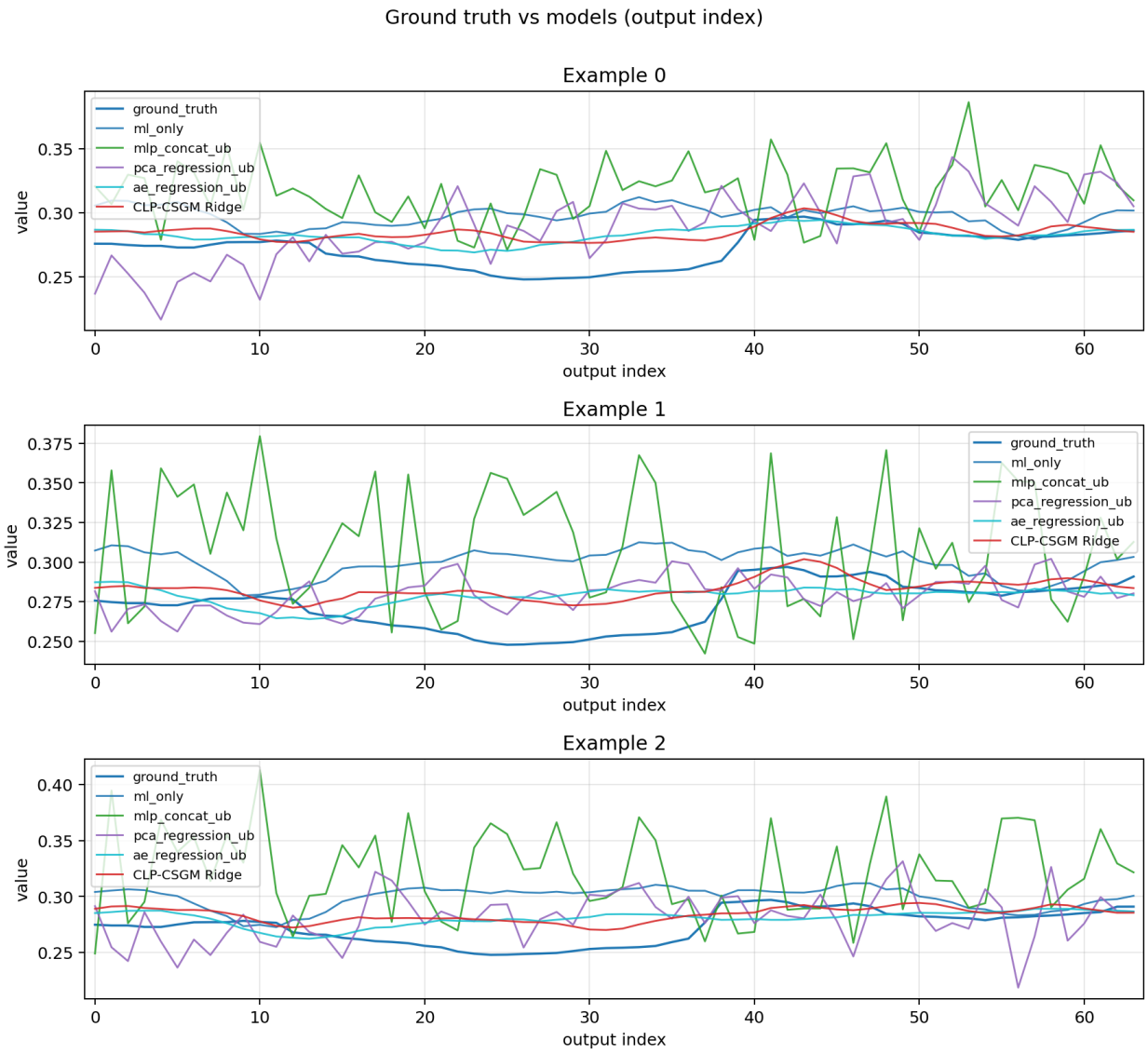


Figure 15: Representative F03-4 test windows. CLP-CSGM Ridge produces smooth reconstructions that follow the ground-truth trend, while direct MLP $[u, b]$ is visibly more oscillatory.

with the harder stress test: the dense input is restricted to GR, and the test block is a contiguous depth interval, not a random split. The trend across measurement ratios remains coherent. Larger ρ generally reduces the full-model test error, and the selected λ remains stable across ratios. The per-window RMSE distributions show that CLP-CSGM Ridge improves the central error relative to both ablated variants while remaining close to AE $[u, b]$, matching the main F03 regime-dependent result.

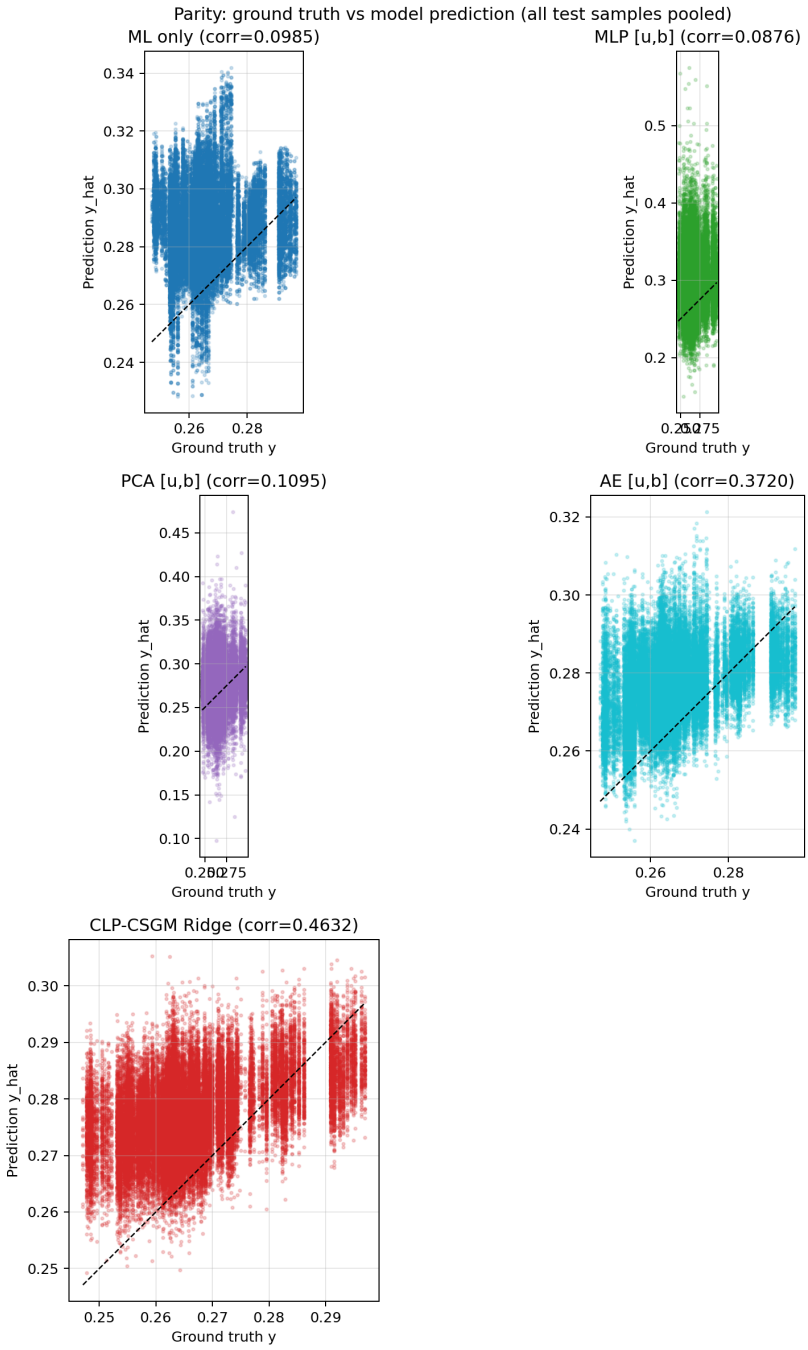


Figure 16: Pooled parity plots for F03-4 test windows. CLP-CSGM Ridge attains the highest displayed correlation among compared methods, while broad scatter remains.

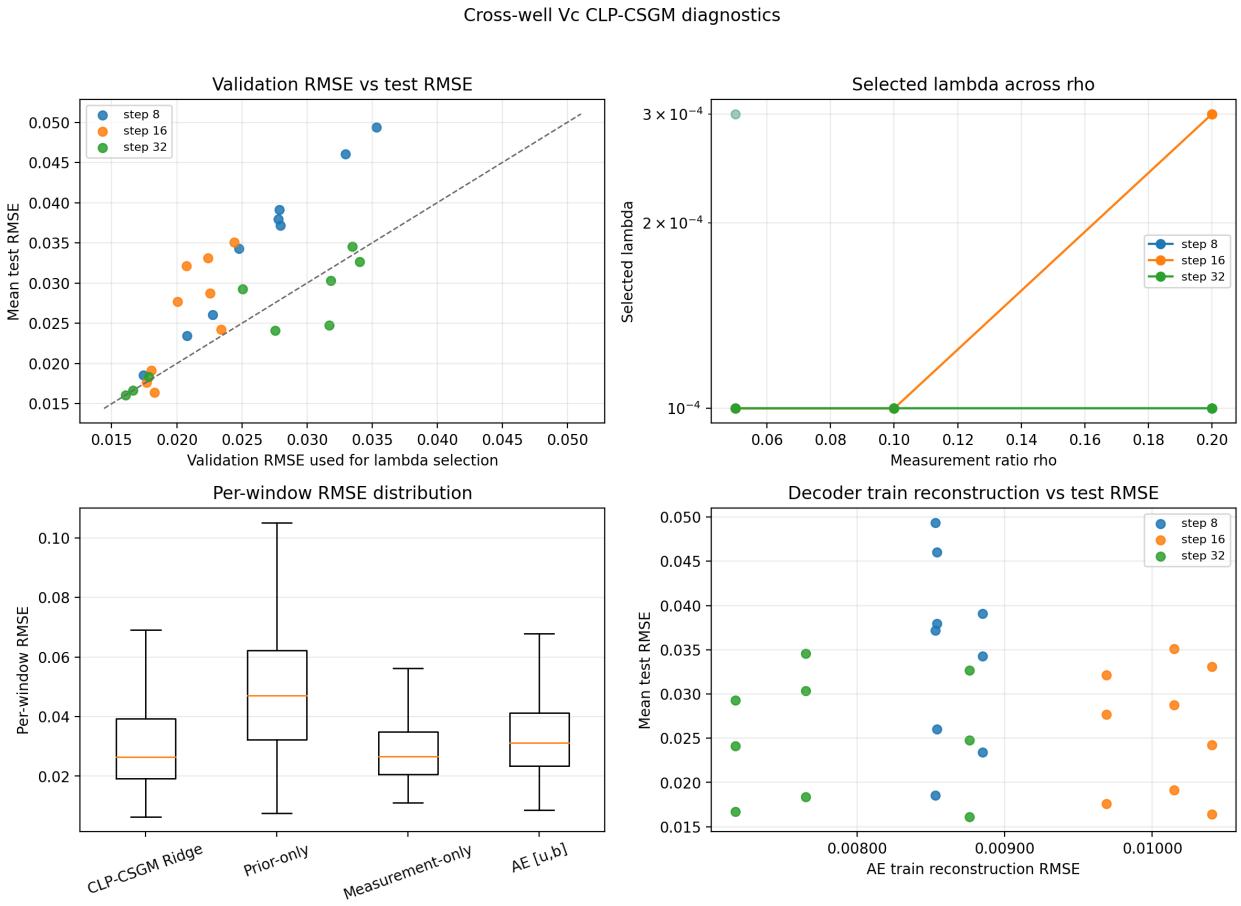


Figure 17: Cross-well V_c diagnostics for CLP-CSGM Ridge summarizing the experiments in Section 6.1. The panels compare validation RMSE with mean test RMSE for CLP-CSGM Ridge, show the selected λ values across measurement ratios, summarize per-window RMSE distributions, and relate autoencoder training reconstruction error to final test RMSE.

B.7 Cross-well Channel-Ablation and Embargo-Style Boundary Check

This subsection reports a paired robustness check for the cross-well V_c step-16 setting. It tests two possible sensitivities: reintroducing GR into u, relative to the main no-GR protocol in Section 5, and removing overlap-prone windows near the train/validation boundary. The latter is an embargo-style boundary check: a small band of windows adjacent to the boundary is removed before validation to create a gap between training and validation windows. We compare four scenarios under the same seeds and measurement-ratio grid, $\rho \in \{0.05, 0.10, 0.20\}$:

- No boundary check + GR in u (stronger dense context reference).
- Embargo-style removal (4 boundary windows) + GR in u.

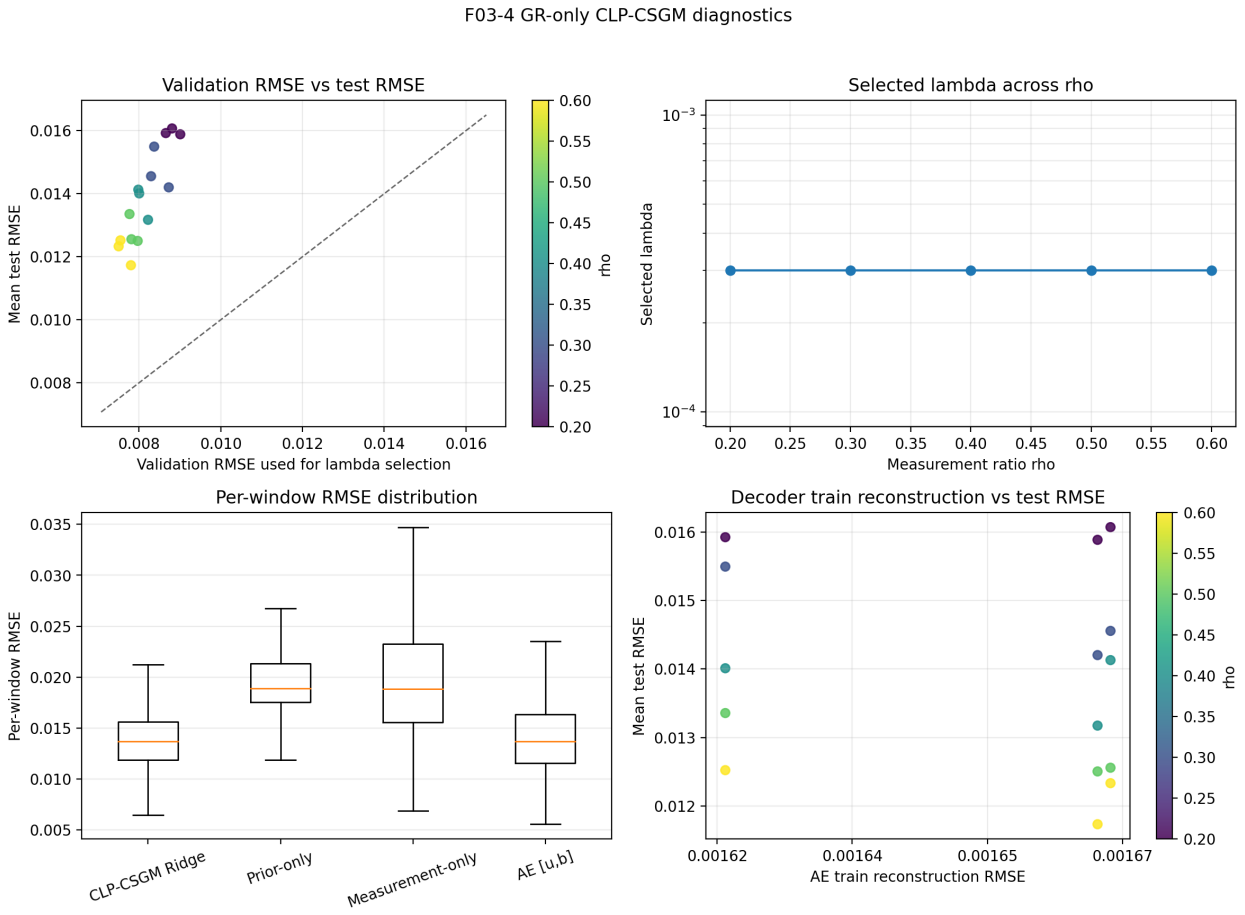


Figure 18: F03-4 GR-only diagnostics for CLP-CSGM Ridge summarizing the experiments in Section 6.2. The panels report validation RMSE versus mean test RMSE, selected λ, per-window RMSE distributions, and the relation between autoencoder training reconstruction error and final test RMSE.

- No boundary check + GR excluded from u (matches the main cross-well u).
- Embargo-style removal (4 boundary windows) + GR excluded from u.

Operationally, the embargo-style removal drops four overlap-prone boundary windows before the validation tail in each training well. In this setup, n_{train} decreases from 52 to 41, while n_{val} and n_{test} stay fixed.

Table 14 shows three stable patterns. Within each boundary protocol, including GR in u lowers absolute RMSE, as expected because V_c is Larionov-derived from GR. Under the no-GR protocol matching the main benchmark, CLP-CSGM Ridge has the lowest RMSE relative to AE $[u, b]$ and measurement-only CSGM throughout the tested grid. The embargo-style boundary check changes the absolute values without reversing this ordering. These checks support the main interpretation:

Table 14: Cross-well step-16 robustness check across GR ablation and embargo-style boundary-check scenarios. Ridge RMSE is reported as mean ± std over three seeds. Gains are relative RMSE reductions of CLP-CSGM Ridge against AE $[u, b]$ and measurement-only CSGM.

scenario	rho	ridge rmse	ae rmse	meas-only rmse	ridge gain vs ae (%)	ridge gain vs meas-only (%)
Emb4+GR	0.05	0.01487 ± 0.00035	0.02056	0.03647	27.7	59.2
Emb4+GR	0.10	0.01406 ± 0.00032	0.01953	0.02849	28.0	50.7
Emb4+GR	0.20	0.01282 ± 0.00026	0.01932	0.02167	33.6	40.8
Emb4-noGR	0.05	0.03117 ± 0.00020	0.03781	0.03647	17.6	14.5
Emb4-noGR	0.10	0.02549 ± 0.00268	0.03439	0.02849	25.9	10.6
Emb4-noGR	0.20	0.01625 ± 0.00139	0.03771	0.02167	56.9	25.0
NoEmb+GR	0.05	0.01403 ± 0.00064	0.01893	0.03622	25.9	61.3
NoEmb+GR	0.10	0.01380 ± 0.00107	0.01802	0.02954	23.4	53.3
NoEmb+GR	0.20	0.01289 ± 0.00052	0.01822	0.02127	29.3	39.4
NoEmb-noGR	0.05	0.03344 ± 0.00151	0.03712	0.03622	9.9	7.7
NoEmb-noGR	0.10	0.02689 ± 0.00235	0.03420	0.02954	21.4	9.0
NoEmb-noGR	0.20	0.01771 ± 0.00138	0.03214	0.02127	44.9	16.7

the cross-well V_c benchmark tests sparse target-log reconstruction under distribution shift without a GR shortcut in u, and the reported gains are not solely an artifact of train/validation boundary overlap.

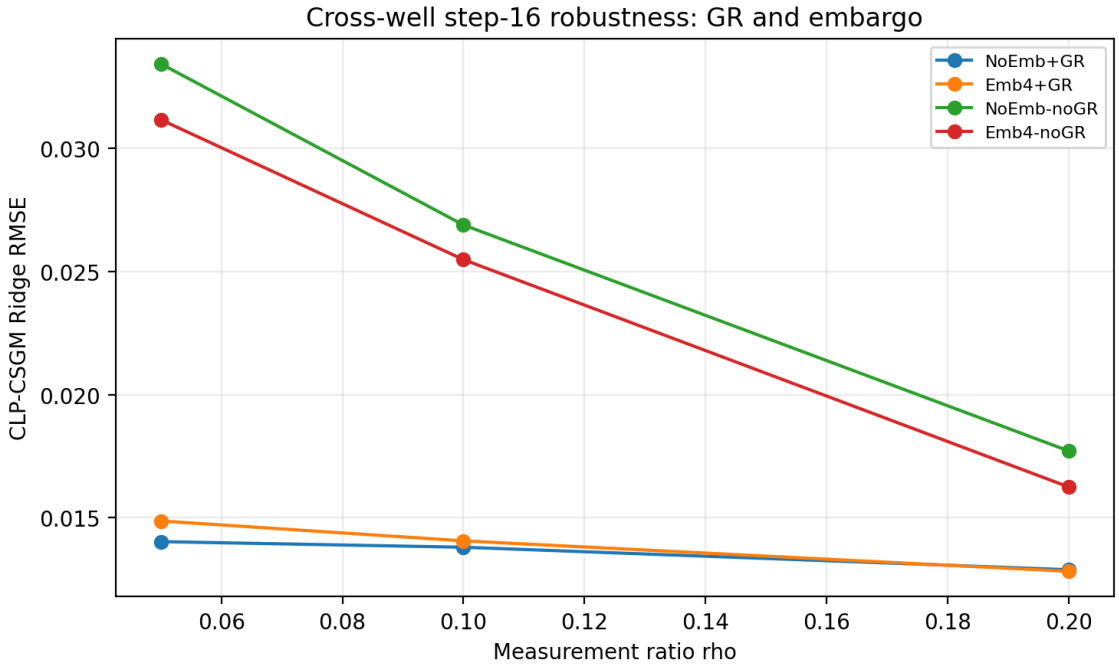


Figure 19: Cross-well step-16 RMSE sensitivity for CLP-CSGM Ridge across GR inclusion in u and embargo-style boundary-check scenarios. Omitting GR from u relative to GR-inclusive rows increases error; the boundary check shifts absolute values without overturning the ranking trends.

B.8 Step-32 development-split resampling

The main cross-well grid uses a depth-tail validation split. At step 32 the development pool is small ($n_{\text{train}} = 27$, $n_{\text{val}} = 3$ after the split). To test whether the CLP-CSGM Ridge versus AE $[u, b]$ gap in that cell depends on the particular validation windows, we redrew the same validation budget ten times inside each development well (`random_per_well`), keeping F03-4 as the held-out test well and freezing the sparse coordinates Ω to the paper seed 7. Prior-only and measurement-only ablations were not rerun.

Table 15: Cross-well step-32 resampling of the development train/validation split ($n = 10$). RMSE is the mean ± standard deviation across splits. Positive mean ΔRMSE (AE − CLP) favors CLP-CSGM Ridge. CLP win frac is the fraction of splits with lower CLP RMSE.

ρ	n_{splits}	n_{train}	n_{val}	CLP RMSE	AE RMSE	mean ΔRMSE	CLP win frac
0.05	10	27	3	0.0342 ± 0.0024	0.0407 ± 0.0025	0.0064	1.00
0.10	10	27	3	0.0305 ± 0.0032	0.0388 ± 0.0015	0.0083	1.00
0.20	10	27	3	0.0188 ± 0.0010	0.0362 ± 0.0064	0.0174	1.00

Table 15 shows that CLP-CSGM Ridge has lower mean RMSE than AE $[u, b]$ at every tested ρ, and lower RMSE in all ten splits. This check does not add a second held-out well and does not vary Ω; it only resamples the small development split.

B.9 Ten-seed CLP versus AE panel

The main cross-well intervals use three seeds (t critical value with df = 2). Several of those paired ΔRMSE intervals include zero, and the CLP-CSGM Ridge versus AE $[u, b]$ ranking changes between step 8 and steps 16–32. We therefore reran that pairwise comparison with ten seeds $\{7, 23, 41, 13, 29, 31, 37, 43, 47, 53\}$, keeping the submitted tail validation split, the no-GR input, and F03-4 as the held-out well. Direct MLP and PCA, structural ablations, and the F03-4 grid were not rerun.

Table 16 shows that none of the nine ten-seed intervals include zero. The CLP-CSGM Ridge advantage is concentrated where training support is smallest (steps 16 and 32, approximately 52 and 27 windows): lower RMSE at every ρ. At step 8, where the development pool is largest (approximately 104 windows), AE $[u, b]$ has lower RMSE at $\rho = 0.05$ and $\rho = 0.10$; CLP-CSGM Ridge has lower RMSE only at $\rho = 0.20$. This is a low-training-support pattern, not a uniform gain at every small measurement ratio. The submitted three-seed tables are unchanged. Each of

Table 16: Cross-well paired $\Delta\text{RMSE} = \text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{CLP}}$ on ten seeds. SE is the seed-level standard error; the 95% half-width uses Student-t with df = 9. Positive mean ΔRMSE favors CLP-CSGM Ridge. Wins count seeds with strictly lower RMSE.

step	ρ	n	mean ΔRMSE	SE	t 95% half-width	CLP wins	AE wins	CI includes 0
8	0.05	10	-0.00826	0.00121	0.00273	0	10	no
8	0.10	10	-0.00318	0.00092	0.00209	2	8	no
8	0.20	10	0.00360	0.00129	0.00293	7	3	no
16	0.05	10	0.00434	0.00159	0.00360	9	1	no
16	0.10	10	0.00677	0.00134	0.00304	10	0	no
16	0.20	10	0.01382	0.00145	0.00328	10	0	no
32	0.05	10	0.00865	0.00129	0.00292	10	0	no
32	0.10	10	0.01217	0.00118	0.00266	10	0	no
32	0.20	10	0.01934	0.00185	0.00418	10	0	no

these seeds also draws its own coordinate set, so the panel does not separate training randomness from observation placement. Appendix B.11 holds the model seeds fixed and redraws only that coordinate set.

B.10 Explicit-mask direct AE

The main direct baselines receive $[\boldsymbol{u}, \boldsymbol{b}]$ with positional identity implicit in the column order. CLP-CSGM instead applies M_Ω to the length-N window. To test whether the CLP-CSGM gain is only that coordinate placement, we fit one additional direct AE on the same nine cells, seeds, and noise. Its input is $[\boldsymbol{u}, \boldsymbol{b}_\Omega, \text{mask}]$: $\boldsymbol{b}_\Omega \in \mathbb{R}^N$ holds the noisy measurement at each observed coordinate and zero elsewhere, and the mask is the binary indicator of Ω. MLP, PCA, ablations, and the F03-4 grid were not rerun, and this method is not added to the main tables.

Table 17: Cross-well CLP-CSGM Ridge versus AE $[\boldsymbol{u}, \boldsymbol{b}]$ and an AE that also receives an explicit window mask ($n = 3$ seeds). Δ is the other method's RMSE minus CLP RMSE. Positive Δ favors CLP-CSGM Ridge. Wins count seeds with strictly lower CLP RMSE.

step	ρ	CLP	AE $[u, b]$	AE mask	Δ AE	Δ mask	CLP>AE	CLP>mask
8	0.05	0.0448	0.0357	0.0340	-0.0092	-0.0108	0/3	0/3
8	0.10	0.0365	0.0320	0.0314	-0.0045	-0.0051	1/3	0/3
8	0.20	0.0227	0.0260	0.0258	0.0033	0.0032	2/3	2/3
16	0.05	0.0334	0.0371	0.0352	0.0037	0.0018	3/3	2/3
16	0.10	0.0269	0.0342	0.0340	0.0073	0.0071	3/3	3/3
16	0.20	0.0177	0.0321	0.0277	0.0144	0.0100	3/3	3/3
32	0.05	0.0322	0.0390	0.0426	0.0068	0.0105	3/3	3/3
32	0.10	0.0264	0.0389	0.0386	0.0125	0.0122	3/3	3/3
32	0.20	0.0170	0.0396	0.0366	0.0225	0.0196	3/3	3/3

Table 17 shows that the explicit mask does not remove the CLP-CSGM advantage where training

support is smallest. At steps 16 and 32, CLP-CSGM Ridge has lower mean RMSE than the masked AE at every ρ. At step 8, the masked AE remains better at $\rho = 0.05$ and $\rho = 0.10$, as the unmasked AE does. The mask shifts the direct AE only modestly and not always downward: at step 32, $\rho = 0.05$, the masked AE has higher mean RMSE than AE $[\boldsymbol{u}, \boldsymbol{b}]$. This check keeps Ω fixed inside each cell; it does not test sensitivity to observation placement. That placement check is Appendix B.11.

B.11 Sparse-coordinate placement

The main protocol samples one coordinate set Ω for each model seed and measurement ratio and then holds it fixed across windows. To test whether the CLP-CSGM Ridge versus AE $[\boldsymbol{u}, \boldsymbol{b}]$ ranking depends on those locations, we redrew Ω five times on the cross-well V_c grid. The model seeds stay at $\{7, 23, 41\}$, so the decoder and the ridge anchor keep the same initialization. The first draw is the published coordinate set. The other four draws change only the generator that samples Ω and the noise on $\boldsymbol{b}$. The direct AE is fit again on each draw, because $\boldsymbol{b}$ is part of its input, and the CLP-CSGM penalty weight is selected again on the validation windows of that draw. The score is the same full-window RMSE as the main grid. Direct MLP, PCA, structural ablations, and the F03-4 grid were not rerun, and these draws are not merged into the submitted three-seed tables.

Table 18: Cross-well CLP-CSGM Ridge versus AE $[\boldsymbol{u}, \boldsymbol{b}]$ across five sparse-coordinate placements and the three paper seeds ($n = 15$ per cell). RMSE values are means over those comparisons. Positive mean ΔRMSE (AE − CLP) favors CLP-CSGM Ridge. Wins count comparisons with strictly lower RMSE.

step	ρ	n	CLP RMSE	AE RMSE	mean ΔRMSE	CLP wins	AE wins
8	0.05	15	0.04231	0.03586	-0.00645	0	15
8	0.10	15	0.03581	0.03225	-0.00356	5	10
8	0.20	15	0.02383	0.02591	0.00208	11	4
16	0.05	15	0.03072	0.03710	0.00638	15	0
16	0.10	15	0.02615	0.03412	0.00797	15	0
16	0.20	15	0.01840	0.03036	0.01196	15	0
32	0.05	15	0.02895	0.03873	0.00977	15	0
32	0.10	15	0.02470	0.03870	0.01401	15	0
32	0.20	15	0.01795	0.04023	0.02228	15	0

Table 18 shows that the cell-level winner matches the published ranking in all nine cells. At steps 16 and 32, CLP-CSGM Ridge has lower RMSE in all fifteen comparisons at every ρ. At step 8 and $\rho = 0.05$, AE $[\boldsymbol{u}, \boldsymbol{b}]$ is lower in all fifteen. The two closer step-8 cells are not unanimous: five

of fifteen comparisons favor CLP-CSGM Ridge at $\rho = 0.10$, and four of fifteen favor the direct AE at $\rho = 0.20$. The low-training-support advantage therefore does not depend on the one published coordinate set. The step-8 ranking is the less stable of the two regimes, and a single extra draw can reverse an individual seed there. This check does not add seeds beyond $\{7, 23, 41\}$.

B.12 Direct linear maps, depth interpolation, and measurement noise

The main grid does not include a direct linear map from $[\boldsymbol{u}, \boldsymbol{b}]$ or an interpolation of the sparse measurements. This supplement adds both on the cross-well V_c grid, with the same windows, seeds $\{7, 23, 41\}$, measurement ratios, and published coordinate draws. Ordinary least squares and ridge regression map standardized $[\boldsymbol{u}, \boldsymbol{b}]$ to the window. The ridge penalty is chosen on validation full-window RMSE. Interpolation connects the noisy measurements along the depth index of the window and keeps the nearest measured value at the ends. It does not use the dense logs. The score is the same full-window RMSE as the main grid. CLP-CSGM Ridge and AE $[\boldsymbol{u}, \boldsymbol{b}]$ in Table 19 are the published offset-0 runs. These comparators are not merged into the submitted tables.

Table 19: Cross-well full-window RMSE at the published noise standard deviation 0.01. Values are means over seeds $\{7, 23, 41\}$. Linear $[u, b]$ is ordinary least squares and Ridge direct is validation-selected ridge, both from $[\boldsymbol{u}, \boldsymbol{b}]$ to the window. Interpolation uses only the sparse measurements. The lowest value in every cell is interpolation.

step	ρ	Linear $[u, b]$	Ridge direct	Interpolation	CLP-CSGM	AE $[u, b]$
8	0.05	0.03122	0.02916	0.01779	0.04483	0.03567
8	0.10	0.02790	0.02384	0.01537	0.03646	0.03195
8	0.20	0.02158	0.01862	0.01100	0.02267	0.02600
16	0.05	0.02903	0.02910	0.01751	0.03344	0.03712
16	0.10	0.02634	0.02661	0.01478	0.02689	0.03420
16	0.20	0.02062	0.02089	0.01115	0.01771	0.03214
32	0.05	0.03382	0.03980	0.01755	0.03217	0.03901
32	0.10	0.03083	0.03878	0.01533	0.02640	0.03894
32	0.20	0.02421	0.02648	0.01125	0.01705	0.03959

Table 19 shows that interpolation has the lowest RMSE in all nine cells. Seed by seed, it is also lowest in all 27 comparisons. CLP-CSGM Ridge is lower than both direct linear maps at step 16, $\rho = 0.20$, and at step 32 for every ρ. At step 8, and at step 16 for $\rho = 0.05$ and $\rho = 0.10$, one or both linear maps are lower than CLP-CSGM Ridge. The measurements at this noise level are a slightly perturbed copy of the same smooth curve, so connecting them is enough.

The same windows and the same standard-normal draws are then rescaled to noise standard

deviations 0.05, 0.10, and 0.20. Table 20 refits CLP-CSGM at each level so that every noise shares one decoder per seed. The penalty grid used for that refit keeps the published values and adds larger weights, up to 100, so validation can reduce the pull of a noisy measurement. At 0.01 and at 0.05 every selected weight still lies inside the published grid. The held-out curve has standard deviation 0.035. The window predicted from the dense logs alone, with no measurement term, has mean RMSE 0.0564 at step 8, 0.0426 at step 16, and 0.0432 at step 32.

Table 20: Interpolation versus CLP-CSGM Ridge when only the measurement-noise standard deviation changes. RMSE values are means over the three paper seeds. Wins count seeds in which CLP-CSGM Ridge has strictly lower RMSE. The held-out V_c standard deviation is 0.035. Noise levels 0.10 and 0.20 are a sensitivity check. At 0.20 the noise is several times that variation, and some selected penalty weights lie above the published grid.

		$\sigma_\eta = 0.01$			$\sigma_\eta = 0.05$			$\sigma_\eta = 0.10$			$\sigma_\eta = 0.20$		
step	ρ	Interp.	CLP	wins	Interp.	CLP	wins	Interp.	CLP	wins	Interp.	CLP	wins
8	0.05	0.0178	0.0456	0	0.0456	0.0508	1	0.0864	0.0547	3	0.1703	0.0557	3
8	0.10	0.0154	0.0365	0	0.0442	0.0456	1	0.0853	0.0539	3	0.1690	0.0555	3
8	0.20	0.0110	0.0212	0	0.0423	0.0372	3	0.0838	0.0513	3	0.1673	0.0548	3
16	0.05	0.0175	0.0334	0	0.0442	0.0386	3	0.0838	0.0416	3	0.1653	0.0424	3
16	0.10	0.0148	0.0269	0	0.0427	0.0352	3	0.0825	0.0416	3	0.1636	0.0424	3
16	0.20	0.0112	0.0159	0	0.0428	0.0286	3	0.0847	0.0387	3	0.1690	0.0424	3
32	0.05	0.0175	0.0317	0	0.0430	0.0382	3	0.0820	0.0423	3	0.1625	0.0430	3
32	0.10	0.0153	0.0264	0	0.0445	0.0365	3	0.0860	0.0414	3	0.1706	0.0431	3
32	0.20	0.0112	0.0170	0	0.0425	0.0279	3	0.0840	0.0381	3	0.1674	0.0409	3

At noise standard deviation 0.05, CLP-CSGM Ridge is lower in every seed at steps 16 and 32 and at step 8 for $\rho = 0.20$. At step 8 for $\rho = 0.05$ and $\rho = 0.10$ the two means remain close, and interpolation still has the lower mean. At 0.10 and 0.20, CLP-CSGM Ridge is lower in all nine cells and in all three seeds. The interpolation advantage is tied to the noise level: it holds at 0.01 and, once the noise reaches the variation of the curve, it no longer holds in the few-window cells. The interpolation error grows because the interpolated curve follows the measurements. The CLP-CSGM error moves less and stays near the dense-log window. Fig. 20 shows one such interval. The two panels use separate horizontal scales. Panel (a) is the published noise. Panel (b) is noise standard deviation 0.05, not the larger sensitivity levels.

On the same splits, the retrospective decoder reconstruction $G(E(\boldsymbol{y}))$ has mean test RMSE 0.0073 at step 8, 0.0099 at step 16, and 0.0110 at step 32. That reconstruction is given the full target window, so it is a check that the decoder can represent the curve. It is not a predictor and it is not ranked with the methods above.

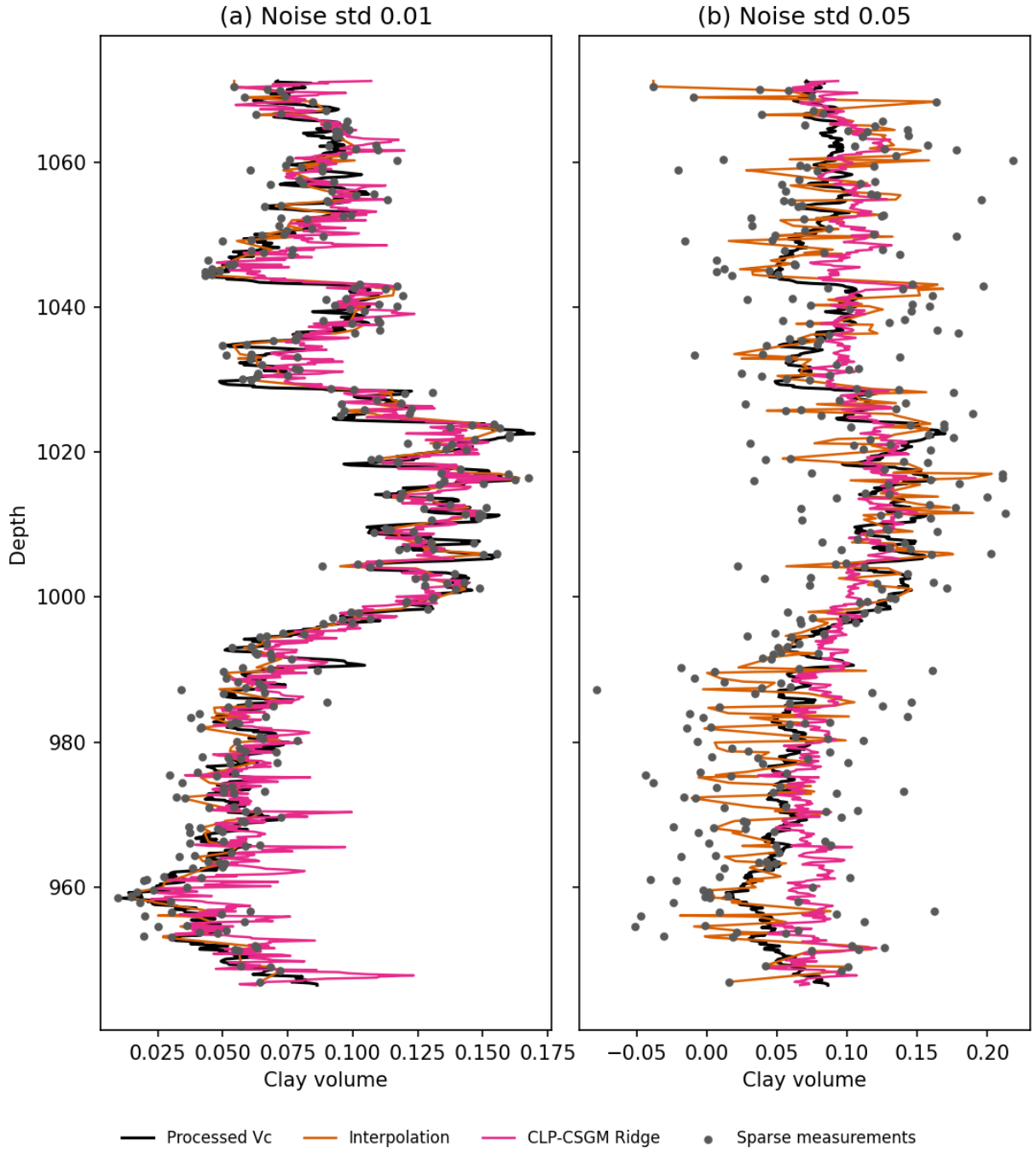


Figure 20: Held-out F03-4 depth tracks, seed 7, step 32, $\rho = 0.20$. Panel (a) uses the published noise standard deviation 0.01. Panel (b) uses 0.05, the same order as the curve's standard deviation 0.035. Each panel has its own clay-volume scale. This interval is one example. Tables 19 and 20 are the comparison.

The step-8 exception at noise standard deviation 0.05 comes from the dense-log map, not from that decoder check. At step 8 the decoder represents the held-out curve with mean test RMSE 0.0073. The window predicted from the dense logs alone has mean RMSE 0.0564. Interpolation at the same step and noise has mean RMSE 0.0456 for $\rho = 0.05$ and 0.0442 for $\rho = 0.10$. Those ratios observe 4 and 6 coordinates in a window of length 64. CLP-CSGM Ridge then stays between the two, at 0.0508 and 0.0456, so the search does not close the gap. At $\rho = 0.20$ the same window observes 13 coordinates, the search does close it, and CLP-CSGM Ridge has mean RMSE 0.0372 against an interpolation RMSE of 0.0423. At steps 16 and 32 the dense-log window itself has mean RMSE 0.0426 and 0.0432, already near the interpolation error at this noise, so the same sparse measurements are enough. The step-8 result is a limit of the dense-log map on this held-out well. It agrees with the ablation in Section 6.3, where that starting point is not helpful at step 8.

B.13 Computational Trade-Offs

Table 21 and Fig. 21 quantify the practical computational trade-off. CLP-CSGM Ridge is more expensive than AE $[u, b]$ because it solves a latent inverse problem at validation and test time instead of producing a single direct forward prediction. Table 21 reports both mean wall-clock time per test window and a separate normalized runtime diagnostic relative to AE; the latter is the diagnostic shown in Fig. 21; it is not the exact quotient of the averaged seconds in the previous column. In the representative cross-well case, the additional cost is substantial relative to AE, but it comes with the lowest mean RMSE among the compared methods. In the F03-4 GR-only case, CLP-CSGM Ridge and AE $[u, b]$ have closer average RMSE, so the runtime gap is less clearly justified. The extra optimization is most readily justified when explicit sparse-measurement consistency produces a meaningful reconstruction gain, especially under stronger ill-posedness or weaker direct supervision.

C Supplementary F03-4 GR-only stress tests

The main F03-4 benchmark in Section 6 uses $\rho \in \{0.20, 0.30, \ldots, 0.60\}$. This appendix summarizes controlled supplements on the same well with GR-only u, L=64, depth-first splits before windowing, coordinate subsampling, the same three fixed random-seed realizations, and the same paired measurement protocol as Section 5. To align the grid with the sparse end of the cross-well study,

Table 21: Representative computational-cost summary. Time is reported as average wall-clock seconds per test window across the tested measurement ratios. The normalized runtime diagnostic column is computed relative to AE $[u, b]$ within each dataset and should not be read as the exact quotient of the averaged wall-clock times.

Dataset	Method	Time/window (s)	Normalized runtime diagnostic vs. AE	Mean RMSE
Cross-well V_c	ML only	0.00787	0.32	0.02582
Cross-well V_c	MLP $[u, b]$	0.04348	2.33	0.21399
Cross-well V_c	PCA $[u, b]$	0.07129	3.15	0.14464
Cross-well V_c	AE $[u, b]$	0.03156	1.00	0.01793
Cross-well V_c	CLP-CSGM Ridge	0.23517	9.79	0.01480
F03-4 GR-only	ML only	0.06010	2.13	0.02391
F03-4 GR-only	MLP $[u, b]$	0.04771	1.73	0.03981
F03-4 GR-only	PCA $[u, b]$	0.31793	11.67	0.02149
F03-4 GR-only	AE $[u, b]$	0.02999	1.00	0.01429
F03-4 GR-only	CLP-CSGM Ridge	0.08838	3.23	0.01418

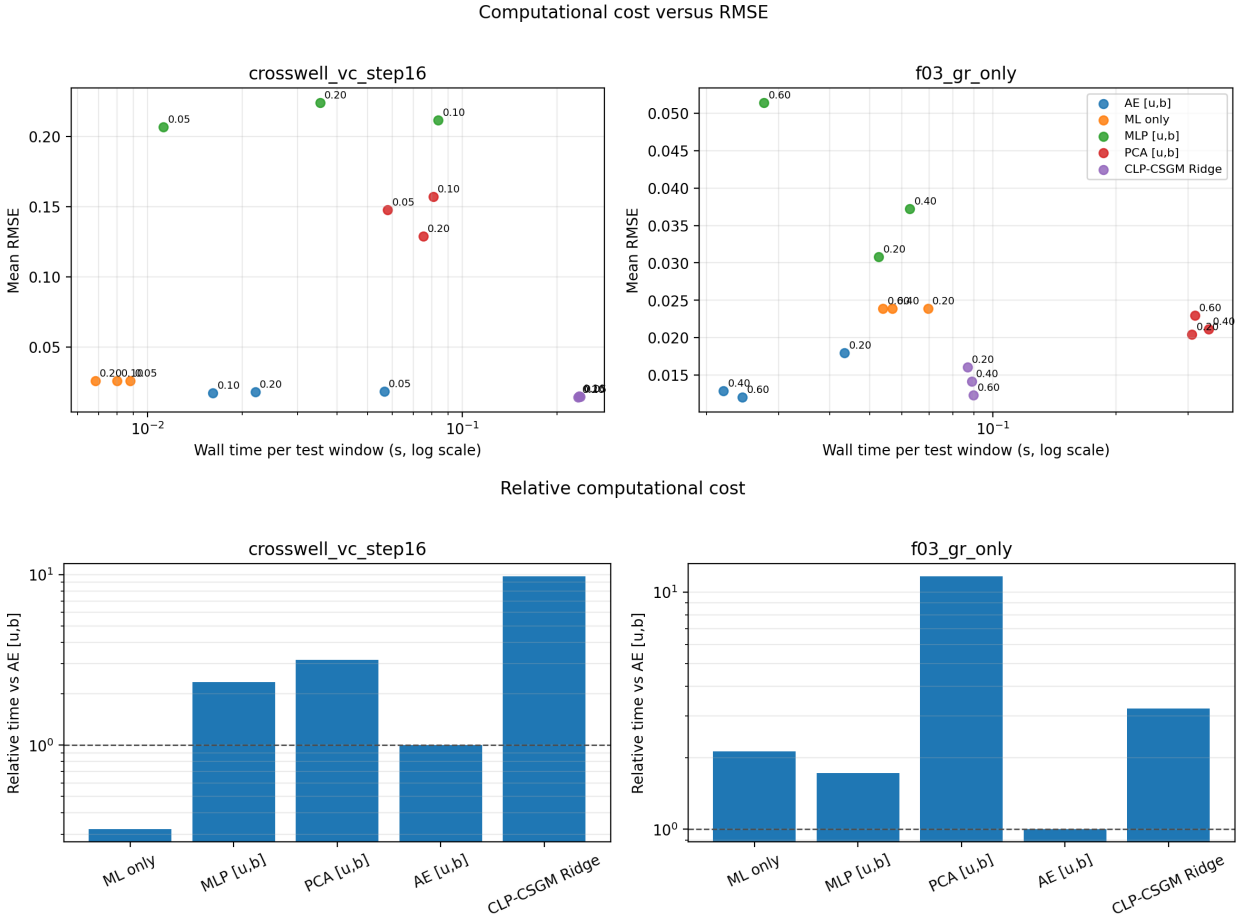


Figure 21: Representative computational-cost diagnostics. The upper panel compares mean RMSE with wall-clock time per test window, and the lower panel reports the normalized runtime diagnostic relative to AE $[u, b]$ within each dataset.

we use $\rho \in \{0.05, 0.10, 0.20\}$ and compare CLP-CSGM Ridge with AE $[u, b]$.

Cases A–D differ only by the stress axes enumerated below:

- **A (baseline noise):** η=0.01 on b=$My + \eta$; standard depth fractions 0.6/0.2/0.2; step 1.
- **B (noisy b):** η=0.05; otherwise as A.
- **C (smaller train pool):** depth fractions 0.45/0.25/0.30; η=0.01; step 1.
- **D (coarse sliding step):** step 4 with standard fractions; η=0.01. This yields far fewer test windows (n_{test}=56 versus 224 in A/B), yielding high-variance paired RMSE summaries at each ρ.

Table 22: F03-4 GR-only stress tests: mean RMSE across seeds for AE $[u, b]$ versus CLP-CSGM Ridge. Relative RMSE gap is $(\text{RMSE}_{\text{AE}} - \text{RMSE}_{\text{Ridge}})/\text{RMSE}_{\text{AE}}$ in percent (positive favors Ridge). Test counts n_{test} count windows in the test depth segment after sliding-window construction.

Case	ρ	AE $[u, b]$	CLP-CSGM Ridge	Rel. RMSE gap (%)	n_{test}
A	0.05	0.02011	0.01682	+16.36	224
	0.10	0.01900	0.01596	+15.98	224
	0.20	0.01489	0.01314	+11.74	224
B	0.05	0.02444	0.01919	+21.51	224
	0.10	0.02397	0.01897	+20.87	224
	0.20	0.02125	0.01846	+13.13	224
C	0.05	0.01810	0.01596	+11.84	369
	0.10	0.01712	0.01509	+11.85	369
	0.20	0.01433	0.01243	+13.26	369
D	0.05	0.01676	0.01672	+0.24	56
	0.10	0.01493	0.01547	-3.62	56
	0.20	0.01115	0.01298	-16.33	56

In A–C, CLP-CSGM Ridge has lower mean RMSE than AE $[u, b]$ at every tested ρ, including under larger η and smaller training fractions. In D, mean RMSE crosses over at $\rho \in \{0.10, 0.20\}$ and is similar to or slightly lower at ρ=0.05. The small effective test count is consistent with this behavior because aggressive overlap reduction amplifies seed variability. The A–D results position the coarse-step F03 stress as an exploratory supplement to the primary ρ sweep in Table 6.

D Supplementary Lapa Pre-salt Permeability Case Study

This appendix reports a complementary case study built from the well-log and permeability data context described by Alvarez et al. [2025] for the Lapa pre-salt carbonate reservoir in the Santos

Basin. Geologically, the Lapa oilfield targets the Aptian Barra Velha Formation, a heterogeneous pre-salt carbonate succession in which laminated microbialites, spherulite, stromatolite, rudstone, and grainstone facies coexist. In such rocks, permeability is controlled by pore structure and diagenesis as much as by porosity. Permeability is therefore the sparse reconstruction target in this supplementary check. Full core–log integration, hydraulic flow units, and nonlinear permeability prediction details are reported in Alvarez et al. [2025]. The benchmark adaptation uses the same CLP-CSGM protocol as the main paper: sliding windows, contiguous train/validation/test split, coordinate subsampling for sparse target observations, paired seeds and measurement ratios, and matched direct $[u, b]$ baselines.

This limited-sample case provides supplementary evidence. The configuration uses $L = 16$, step $= 4$, seven dense channels in u, target k_{lab} in $\log_{10}$ scale, $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$, and the same three fixed random-seed realizations used only for reproducible paired averaging.

Table 23 shows that both CLP-CSGM prior variants remain competitive with direct $[u, b]$ baselines across the tested ρ grid. CLP-CSGM MLP and CLP-CSGM Ridge have lower mean RMSE than AE $[u, b]$, PCA $[u, b]$, ML only, and direct MLP $[u, b]$ at all reported ratios. The comparison with measurement-only CSGM is ratio-dependent: measurement-only has lower mean RMSE at some lower or intermediate ratios, while the prior-guided variants have lower mean RMSE at others. This variation is consistent with the expected regime dependence when sparse measurements are informative but the sample size is limited.

E Laboratory porosity on a cored interval

This supplement uses the 87 laboratory-porosity samples of well 861 over 5205.91–5233.72 m. The target is $\mathrm{Phi}_{\mathrm{lab}}$ as recorded at those depths. It is not a porosity curve computed from density or sonic. Dense context is gamma ray, deep resistivity, shallow resistivity, and lithotype. Density, neutron porosity, sonic porosity, and $\mathrm{Phi}_{\mathrm{ND}}$ are excluded because they track $\mathrm{Phi}_{\mathrm{lab}}$.

Depth is split into three contiguous blocks before windowing, so training and test windows do not share rows. Windows have length 8 and step 1. The last 16 rows of the deepest training block form the validation segment. Seeds are $\{7, 23, 41\}$. For each fold, seed, and ratio, one coordinate set is shared by every window. The laboratory value on those coordinates is the measurement,

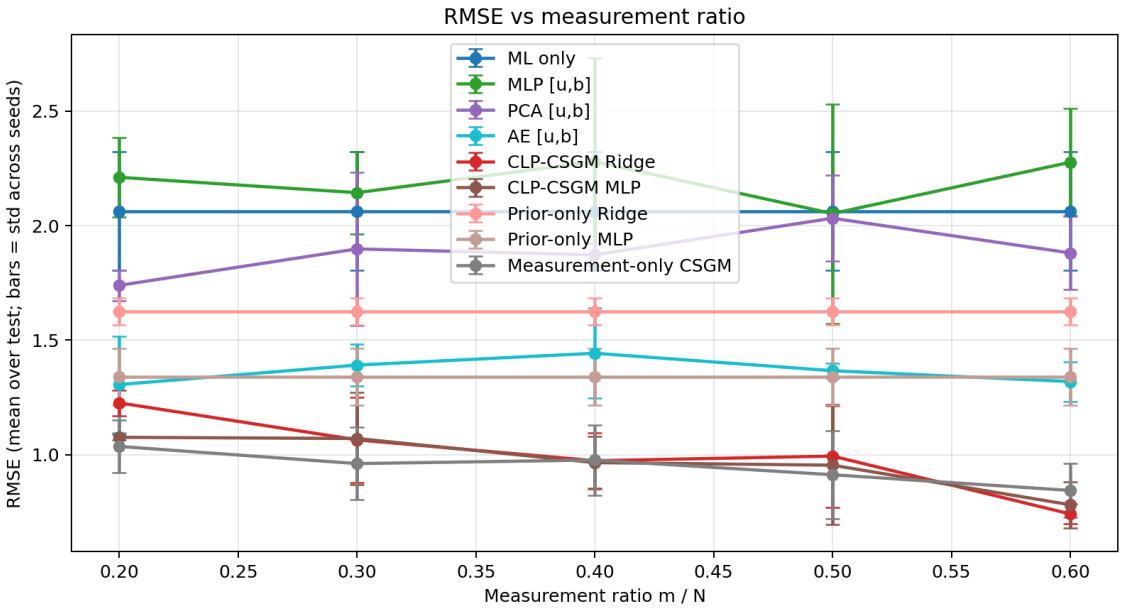


Figure 22: Supplementary Lapa case study RMSE by measurement ratio. Prior-guided CLP-CSGM variants remain competitive across the sparse-measurement grid.

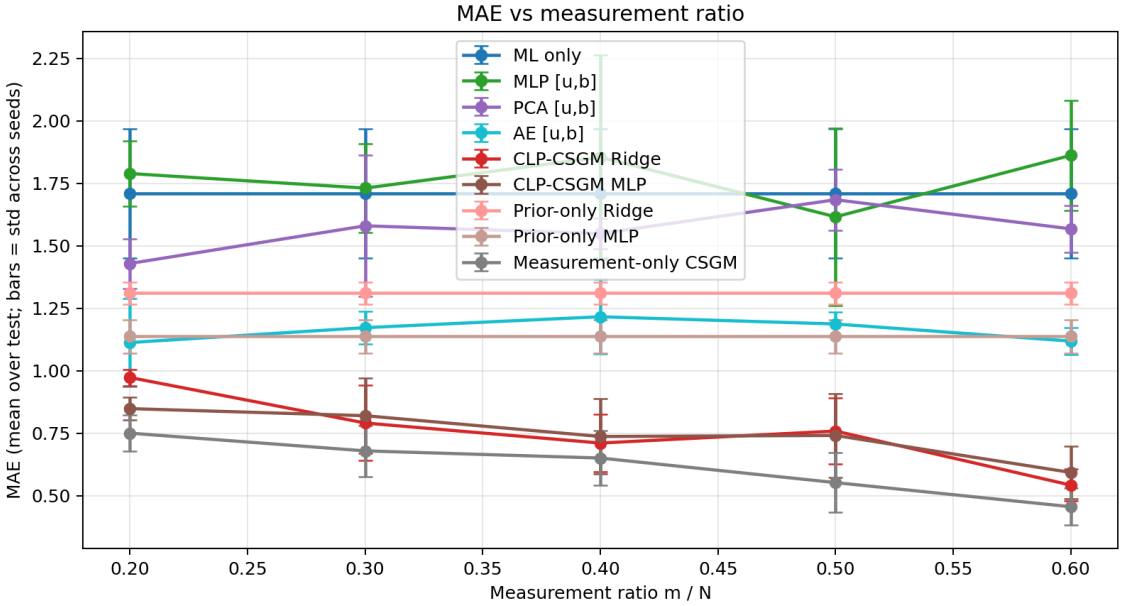


Figure 23: Supplementary Lapa case study MAE by measurement ratio. The ranking pattern is consistent with the RMSE view.

Table 23: Supplementary Lapa case study by measurement ratio. Values are means over three seeds.

rho	method	rmse	mae	relative_l2
0.20000	CLP-CSGM MLP	1.07615	0.84823	1.00530
0.20000	CLP-CSGM Ridge	1.22639	0.97273	1.08660
0.20000	Measurement-only CSGM	1.03628	0.75013	0.93124
0.20000	AE [u,b]	1.30639	1.11253	1.22956
0.20000	PCA [u,b]	1.73942	1.42881	1.60571
0.20000	ML only	2.06280	1.70838	2.00587
0.20000	MLP [u,b]	2.21106	1.78860	2.13801
0.30000	CLP-CSGM MLP	1.07099	0.81987	0.96733
0.30000	CLP-CSGM Ridge	1.06503	0.79065	0.91318
0.30000	Measurement-only CSGM	0.96152	0.67917	0.86494
0.30000	AE [u,b]	1.39133	1.17212	1.30771
0.30000	PCA [u,b]	1.89848	1.57954	1.72206
0.30000	ML only	2.06280	1.70838	2.00587
0.30000	MLP [u,b]	2.14391	1.73034	2.00236
0.40000	CLP-CSGM MLP	0.96555	0.73670	0.88465
0.40000	CLP-CSGM Ridge	0.97383	0.71050	0.86519
0.40000	Measurement-only CSGM	0.97680	0.65052	0.86968
0.40000	AE [u,b]	1.44311	1.21575	1.35981
0.40000	PCA [u,b]	1.87246	1.54857	1.79426
0.40000	ML only	2.06280	1.70838	2.00587
0.40000	MLP [u,b]	2.27913	1.85405	2.18847
0.50000	CLP-CSGM MLP	0.95447	0.74089	0.84011
0.50000	CLP-CSGM Ridge	0.99413	0.75851	0.85865
0.50000	Measurement-only CSGM	0.91277	0.55221	0.80389
0.50000	AE [u,b]	1.36684	1.18683	1.24828
0.50000	PCA [u,b]	2.03242	1.68385	1.85003
0.50000	ML only	2.06280	1.70838	2.00587
0.50000	MLP [u,b]	2.05267	1.61547	2.03388
0.60000	CLP-CSGM MLP	0.78164	0.59283	0.70811
0.60000	CLP-CSGM Ridge	0.74144	0.54218	0.67252
0.60000	Measurement-only CSGM	0.84417	0.45574	0.74392
0.60000	AE [u,b]	1.31942	1.11830	1.22740
0.60000	PCA [u,b]	1.88084	1.56736	1.77341
0.60000	ML only	2.06280	1.70838	2.00587
0.60000	MLP [u,b]	2.27679	1.86123	2.16631

with no added noise. RMSE is computed only on the remaining coordinates. $\rho = 0.25$ uses $m = 2$ measurements and $\rho = 0.50$ uses $m = 4$. The direct AE receives $[\boldsymbol{u}, \boldsymbol{b}]$ with the same coordinates. The log-only anchor is $G(h(\boldsymbol{u}))$ without the measurement term. Neither method is added to the main F3 tables.

Table 24 shows that the direct AE has the lower mean held-out RMSE at both ratios. CLP-CSGM Ridge has the lower RMSE in five of nine splits at $\rho = 0.25$ and in six of nine at $\rho = 0.50$, and it has lower RMSE than the log-only anchor in all nine splits at both ratios. The two summaries answer different questions. The mean asks which method is lower after every split is weighted

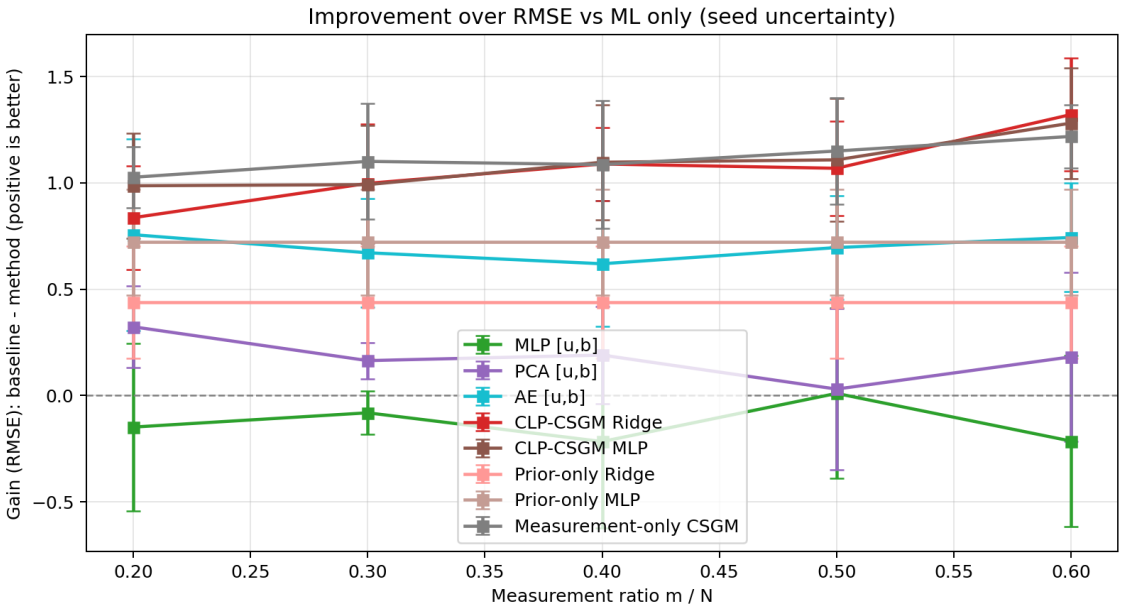


Figure 24: Supplementary Lapa case study RMSE gain over ML-only. Positive values indicate improvement over dense-context-only prediction.

Table 24: Well 861 laboratory-porosity supplement. Entries are the mean held-out RMSE across three depth blocks and three seeds ($n = 9$). Δ is AE minus CLP-CSGM Ridge; a negative value means the direct AE has the lower mean. CLP>AE counts splits in which CLP-CSGM Ridge has the lower RMSE.

ρ	CLP	AE $[u, b]$	prior-only	Δ AE	CLP>AE
0.25	0.0602	0.0537	0.0729	-0.0065	5/9
0.50	0.0580	0.0523	0.0753	-0.0057	6/9

equally. The split count asks how often CLP-CSGM Ridge is the lower of the two. They disagree because one depth block carries errors large enough to set the mean.

Table 25: Well 861 held-out laboratory-porosity RMSE by depth block. Each entry is the mean across the three seeds. Prior-only is $G(h(\boldsymbol{u}))$ with no measurement term.

Block	ρ	CLP	AE $[u, b]$	prior-only
5205.91–5214.48 m	0.25	0.0338	0.0371	0.0486
5205.91–5214.48 m	0.50	0.0228	0.0350	0.0472
5214.67–5223.63 m	0.25	0.0536	0.0548	0.0559
5214.67–5223.63 m	0.50	0.0497	0.0543	0.0554
5224.39–5233.72 m	0.25	0.0931	0.0691	0.1142
5224.39–5233.72 m	0.50	0.1016	0.0676	0.1234

Table 25 locates the tradeoff. On 5205.91–5214.48 m, CLP-CSGM Ridge has the lower block-mean RMSE at both ratios, and raising ρ from 0.25 to 0.50 lowers that RMSE from 0.0338 to 0.0228 while the direct AE stays near 0.035. On 5214.67–5223.63 m the two methods are close,

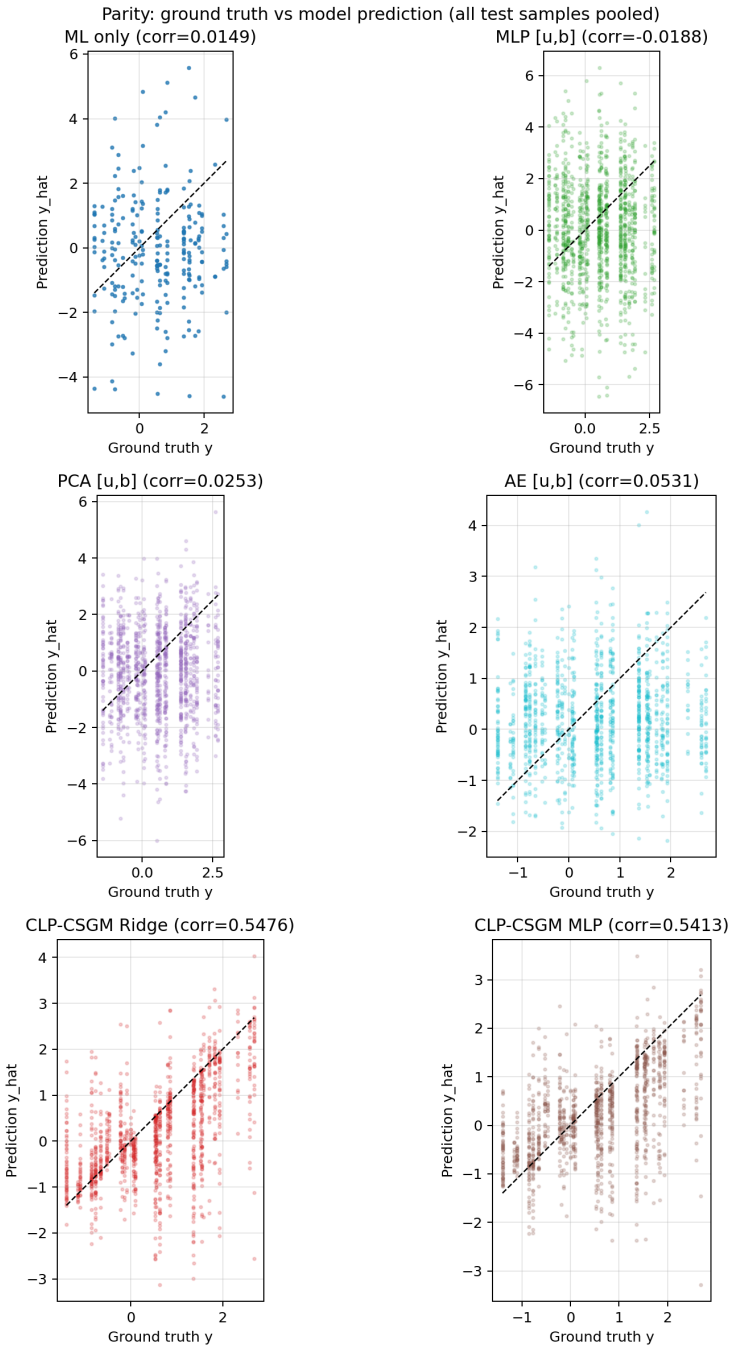


Figure 25: Supplementary Lapa pooled parity plot. The visual agreement is consistent with the quantitative summary, while still reflecting variability expected under small test counts.

with CLP-CSGM Ridge still slightly lower. On 5224.39–5233.72 m the direct AE is lower by a wide margin (0.0691 and 0.0676 against 0.0931 and 0.1016).

The gain where CLP-CSGM Ridge is ahead has two parts. First, the measurement term is doing the work the method is built for: in every block and at both ratios, CLP-CSGM Ridge has lower RMSE than the log-only anchor. The laboratory coordinates pull the latent code off $h(\boldsymbol{u})$. Second, that pull is most useful when the anchor is already a usable start. In the shallowest block, mean $\mathrm{Phi}_{\mathrm{lab}}$ is 0.092 with standard deviation 0.019, mean gamma ray is about 15 API, and the median deep resistivity is about 1900. Adding two further laboratory coordinates ($\rho = 0.50$ rather than 0.25) then cuts the CLP-CSGM Ridge error by about one third. The direct AE already receives those coordinates as input features, so the same increase in ρ changes it little. The explicit residual, not a larger training pool, is what improves CLP-CSGM Ridge there. Each test fold still has 28 to 35 overlapping training windows, so this is not the scarce-window regime of the F3 steps 16 and 32.

The loss on the mean comes from the deepest block, and it is a context shift relative to the two blocks that train the anchor when that block is held out. Mean $\mathrm{Phi}_{\mathrm{lab}}$ rises to 0.145, mean gamma ray rises to about 30 API, and the median deep resistivity falls to about 600. The lithotype mix also changes: codes 2 and 4 account for 25 of the 29 rows, whereas the middle block is mostly code 1. The ridge anchor is a linear map from $\boldsymbol{u}$ fitted on the other blocks, so $\boldsymbol{z}_0 = h(\boldsymbol{u})$ starts far from the laboratory windows. With only $m = 2$ or $m = 4$ measurements in a length-8 window, the residual still improves on that anchor (0.0931 versus 0.1142, and 0.1016 versus 0.1234) but does not catch the direct AE. The AE never has to pass through the shifted anchor: the same $\boldsymbol{b}$ enters as features and the map is applied in one step. Windows overlap inside each block, so the block contrast, not the count of windows, is the depth evidence.

The tradeoff is therefore where the anchor sits. CLP-CSGM Ridge gains when dense logs seen in training still describe the test interval and the laboratory coordinates can finish the correction. It loses the pooled mean when the test logs move outside that support and only a few measurements are available to repair $\boldsymbol{z}_0$. The result does not show that the measurement term fails, and it does not extend the F3 low-support advantage to a mean win on this laboratory target. It is one cored interval.